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**Transformer-Based Semantic Segmentation
for Autonomous Driving
in Adverse Weather Conditions
Using Weather- and Time-Aware Supervision**

Master's Thesis

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Declaration

I hereby declare that I have written this thesis independently and have not used any sources or aids other than those indicated in the references. All passages which are taken from published or unpublished sources are explicitly marked as such. This thesis has not been submitted, in whole or in part, for any other academic degree or examination.

Ingolstadt, 25.12.2025

Shiv Chaudhary

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Abstract

Autonomous vehicles require robust perception that functions reliably under both clear daytime conditions and adverse weather/illumination such as rain, fog, snow, and night. While modern semantic segmentation models like DeepLabV3+ [1] perform well on clear-weather benchmarks like Cityscapes [2], their accuracy degrades significantly on adverse-condition datasets such as ACDC [3]. Typically, models excel in either normal or adverse conditions, but not both, and most specialize in single adverse conditions. This thesis addresses the gap by developing a single semantic segmentation model robust across all five conditions.

We investigate transformer-based segmentation for adverse-weather perception. Starting from Mask2Former [4] with Swin-L backbone [5] pretrained on Cityscapes, we propose a novel weather- and time-aware training strategy using only real-world data. Our method combines three components: (i) alternate-batch training on Cityscapes (clear) and ACDC (adverse), (ii) selective encoder freezing on Cityscapes batches to preserve clear-weather features, and (iii) auxiliary Weather-Aware Supervision (WAS) and Time-Aware Supervision (TAS) heads that encourage domain-sensitive features during training. These auxiliary heads are used only during training; inference uses standard Mask2Former.

Experiments on Cityscapes and ACDC [2, 3] using the 19-class label space systematically evaluate our contributions. During training, the official training splits of Cityscapes and ACDC were each divided into an internal 85% training and 15% validation split using a fixed seed. All reported results are computed on the official validation splits of Cityscapes and ACDC, treated as held-out evaluation sets for this thesis (the Cityscapes test set is not used). We establish baselines by training DeepLabV3+ and Mask2Former on Cityscapes only [1, 4], revealing substantial ACDC performance drops. We then demonstrate progressive gains: first with Cityscapes+ACDC alternate training and encoder freezing, then with WAS/TAS heads. This ablation isolates each component’s impact.

Our full method achieves mean Intersection over Union (mIoU) of 0.83 on Cityscapes and 0.77 on ACDC, an absolute 14-percentage-point improvement on ACDC over Cityscapes-only Mask2Former while maintaining clear-weather performance. Ablation studies show training with both datasets is essential for robustness; WAS/TAS provides modest but consistent improvement, especially for night scenes; alternation schedule and encoder freezing have smaller impact. Results demonstrate a carefully designed training strategy substantially enhances robustness across adverse conditions without compromising clear-weather performance, though with increased computational cost versus CNN baselines.

Resources: [GitHub repository](#) | [Pretrained checkpoints](#)

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List of Acronyms

ACDC	Adverse Conditions Dataset with Correspondences
CS	Cityscapes (clear-weather dataset)
mIoU	Mean Intersection over Union
WAS	Weather-Aware Supervision
TAS	Time-Aware Supervision
CNN	Convolutional Neural Network
ViT	Vision Transformer
GPU	Graphics Processing Unit

1 Introduction

1.1 Background and Motivation

Semantic segmentation provides dense, pixel-wise classification of a scene, assigning every image pixel to a semantic class such as road, car, pedestrian, building, or vegetation [1], [6]. In contrast to object detection, which outputs a set of bounding boxes, semantic segmentation produces a full-resolution label map in which each class is visualised with a distinct colour.

Semantic segmentation for urban driving

For urban driving scenes, semantic segmentation yields a rich representation of:

- the **drivable area** (road, lane boundaries, sidewalks),
- **dynamic traffic participants** (cars, trucks, buses, motorcycles, bicycles, pedestrians, riders),
- **static infrastructure** (buildings, walls, fences, poles, traffic lights, traffic signs),
- **background context** (vegetation, terrain, sky).

This representation is attractive for autonomous driving because it aligns directly with downstream tasks such as free-space estimation, object tracking and risk assessment. In this thesis, all experiments use the standard *19-class* label space defined by the Cityscapes dataset [2]. The same label space is applied to both Cityscapes (clear weather) and ACDC (adverse weather and night-time), so that performance is directly comparable across conditions. Details of the datasets, label space and colour palette are provided in Chapter 3.

Limitations of clear-weather benchmarks

Key benchmark datasets for semantic segmentation in autonomous driving, such as Cityscapes, focus mainly on clear, daytime conditions [2]. Models trained under such conditions typically show strong performance on in-distribution evaluation sets, but their accuracy drops sharply when deployed in adverse conditions such as rain, fog, snow or night-time scenes [3], [7], [8], [9], [10], [11]. In exactly those situations where visibility decreases, contrast is reduced, textures become ambiguous and sensor noise increases, the perception system should remain reliable from a safety perspective. In practice, many existing models struggle the most when they are needed the most.

Recent work has highlighted the severe domain gap between clear-weather datasets such as Cityscapes and adverse-condition datasets such as ACDC [2], [3], [7], [10]. Collecting and

annotating large amounts of high-quality adverse-weather data is significantly more challenging than acquiring clear-weather images, which results in strongly unbalanced datasets. A common workaround in practice is to train two separate models—one for clear conditions and one for adverse conditions—or to fine-tune a clear-weather model on adverse-weather images. Both strategies typically improve performance in the new domain but degrade accuracy on the original one due to domain shift and catastrophic forgetting [3], [7].

Motivation for this thesis

For real-world deployment, relying on multiple specialised models is impractical, and sacrificing clear-weather performance to improve robustness in adverse conditions is undesirable. What is needed is a *single* semantic segmentation model that can handle both clear and adverse conditions reliably.

In contrast to approaches that rely heavily on synthetic data to simulate adverse scenarios [7], this thesis focuses strictly on real-world datasets. Cityscapes (clear daytime) and ACDC (adverse weather and night-time) are used for both training and evaluation [2], [3], and the core question is whether carefully designed training strategies and architectures can reduce the domain gap enough to enable one model that performs well across all conditions.

1.2 Problem Statement

In the context of autonomous driving, a perception system that performs well only in clear daytime conditions is insufficient. Real-world driving involves frequent transitions between weather and illumination states: rain storms, fog banks, wet snow, dusk, night-time under artificial illumination and combinations thereof. If a semantic segmentation model has been trained mainly on clear-weather data, its predictions in adverse conditions can be inconsistent, noisy or systematically biased [3], [7].

Baseline performance breakdown under adverse conditions

To make the problem concrete, it is useful to look at how a standard segmentation model behaves on individual images. For each scene, we can compare three elements: the input RGB image, the manually annotated ground-truth mask (what the segmentation *should* look like), and the predicted mask produced by the model (what the segmentation *actually* is). The visual differences between ground truth and prediction directly reveal where the model struggles.

Figure 1.1 shows such a comparison for typical urban scenes from ACDC (rain, fog, snow, night) and Cityscapes (clear). The predictions are generated by a DeepLabV3+ model that is *trained only on the Cityscapes dataset* and then evaluated on both datasets, following the training protocol of Kerim et al. [7]. This setup represents the common practice of training a single model on a clear-weather benchmark and expecting it to generalise to adverse conditions.

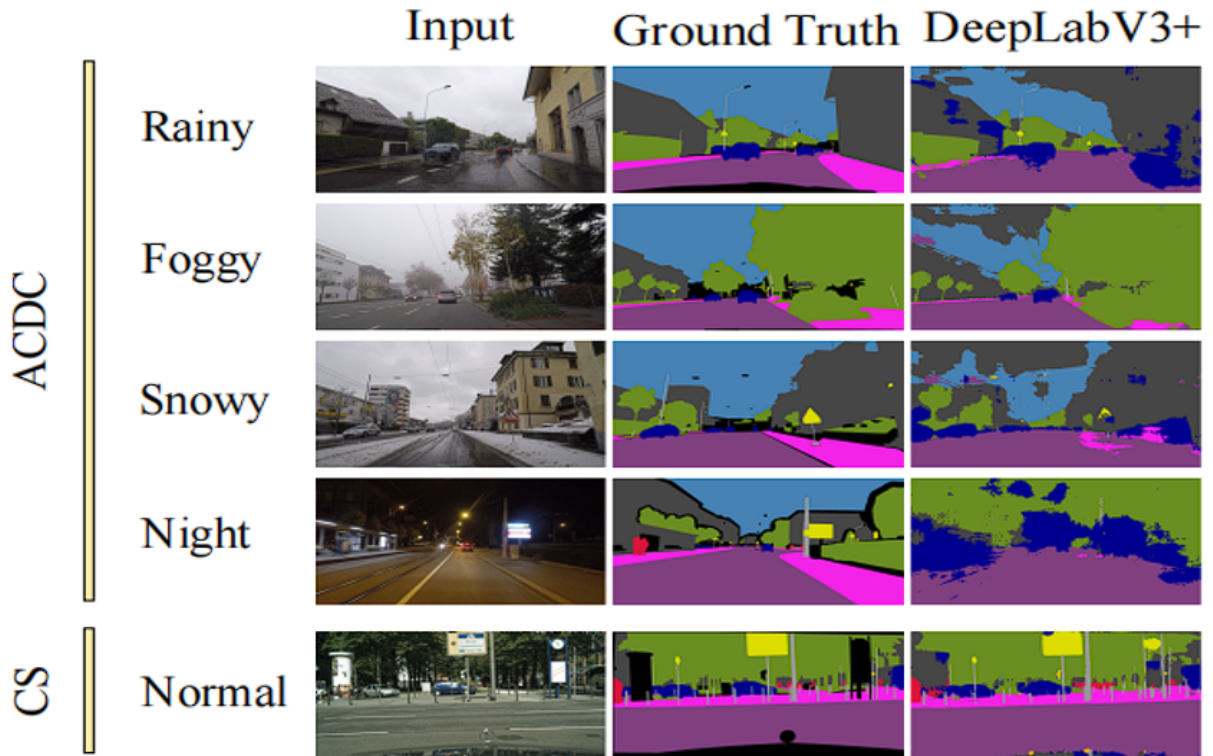


Figure 1.1: Visual comparison between ground-truth segmentation and DeepLabV3+ predictions on ACDC (rain, fog, snow, night) and Cityscapes (clear) scenes. The DeepLabV3+ model is trained only on Cityscapes (clear-weather) images and evaluated on both datasets. It performs well on the normal Cityscapes image but degrades strongly under adverse ACDC conditions, illustrating the domain gap between clear and adverse weather (adapted from Kerim et al. [7]).

On the clear Cityscapes image, the predicted mask is visually very close to the ground truth: road, sidewalk, vehicles and buildings are segmented correctly with only minor deviations. In the adverse ACDC images, however, the same Cityscapes-trained model produces substantial errors: road and sidewalk boundaries are blurred or missing in rain; large parts of the scene are misclassified as vegetation or building in fog; snow-covered regions confuse the model about where the road is; and at night many objects disappear into the background or are segmented as large, amorphous blobs.

1.3 Challenges and Research Gap

The qualitative examples in Figure 1.1 show that a model trained on clear-weather Cityscapes data works well on its training domain but fails under adverse weather. The reverse is also true: a model tuned only on adverse-weather data tends to lose performance on normal conditions. This behaviour is driven by three closely related challenges.

1.3.1 Domain gap between normal and adverse conditions

Cityscapes and ACDC share the same 19-class label space and similar urban driving scenes, but the underlying image distributions are very different. Cityscapes consists almost exclusively of clear daytime images with good visibility and stable illumination [2], whereas ACDC covers heavy rain, dense fog, snow and night-time scenarios with strong illumination changes and sensor artefacts [3]. From the model’s perspective, this constitutes a substantial domain gap: textures, colour histograms, contrast and even the apparent geometry of objects change across domains.

A model trained only on the normal-weather domain will learn features that are highly tuned to that distribution. When such a model is evaluated on ACDC, many of these features no longer activate reliably, leading to the misclassifications and hallucinated objects seen in Figure 1.1. Conversely, a model fine-tuned only on adverse weather adapts to the ACDC appearance but forgets the statistics of clear daytime scenes, degrading again on Cityscapes. This is the classic domain adaptation problem in semantic segmentation [3], [7], [10], but here the goal is not to adapt from a source to a single target, but to support *both* domains with one model.

1.3.2 Complexity of adverse-weather data

Adverse conditions make the segmentation task inherently harder:

- **Rain** introduces streaks, reflections and specular highlights on wet road surfaces, which break the usual visual cues for lane markings and road boundaries.
- **Fog** reduces global contrast and obscures distant objects, so boundaries between sky, buildings and background vegetation become ambiguous.
- **Snow** changes the appearance of the road and sidewalks completely, often turning the entire ground plane into a bright, texture-poor region.
- **Night** brings extreme illumination imbalance: large dark areas combined with very bright headlamps and streetlights, which cause glare, blooming and sensor noise.

Even for human observers, these scenes are harder to interpret than clear daytime images. For a model that has only seen normal-weather data during pretraining, they represent out-of-distribution inputs and expose blind spots in the learned representation. Recent methods such as VBLC [8], FREST [9] or WA2Net [11] confirm that dedicated treatment of adverse conditions is required to obtain reliable performance.

1.3.3 Imbalanced and limited adverse-weather data

Compared to Cityscapes, the number of labelled adverse-weather images is limited. Annotating dense pixel-level masks for rain, fog, snow and night is expensive and time-consuming, so ACDC is much smaller than what might be desirable for fully supervised training [3], [7]. Simply

training a model from scratch on ACDC is unlikely to reach the same level of performance as a model that benefits from large-scale clear-weather pretraining. At the same time, naive joint training on the union of both datasets can cause the model to focus on the majority domain or to oscillate between the two, without converging to a representation that serves both well.

1.3.4 Implications for autonomous driving

For autonomous driving, these challenges are not academic details. A segmentation model that only works reliably in clear daytime scenes is not sufficient for real-world deployment. Vehicles are expected to operate in all weather and illumination conditions encountered on public roads. If semantic segmentation is to be a core component of the perception stack, it must be robust across the full range of conditions; otherwise, the system cannot earn the level of trust and safety required for wider adoption of automated driving functions. Addressing the domain gap and the complexity of adverse-weather data with a *single* model, without maintaining separate models per condition, is therefore a central requirement and the starting point for this thesis.

1.4 Research Objectives and Proposed Approach

Given the challenges above, the overarching objective of this thesis is to design and evaluate a single semantic segmentation model that achieves high performance on both normal (clear) and adverse (rain, fog, snow, night) urban driving conditions, using only real Cityscapes and ACDC data and a shared 19-class label space. The focus is on adapting a strong transformer-based baseline to this multi-domain setting while preserving its clear-weather performance.

To approach this goal, the thesis makes four concrete design choices, which are implemented and evaluated in the remainder of the work.

1) Using a strong transformer-based baseline (Mask2Former + Swin-L). The first choice is architectural. The 2022 baseline by Kerim et al. [7] uses DeepLabV3+ as the core segmentation model [1]. While DeepLabV3+ remains competitive, more recent transformer-based models such as Mask2Former have shown superior performance on Cityscapes and other benchmarks [4], [12]. Mask2Former formulates image segmentation as a mask-classification problem with a transformer decoder and a pixel decoder, and can be used for panoptic, instance and semantic segmentation within a unified architecture.

In this thesis, the base model is a Mask2Former variant with a Swin-L backbone [5], pretrained on Cityscapes for semantic segmentation. This gives a strong clear-weather starting point: the model already achieves state-of-the-art performance on Cityscapes, so the focus is on how to adapt it to adverse conditions without losing its normal-weather accuracy.

2) Alternate-batch training on Cityscapes and ACDC. The second choice concerns how Cityscapes and ACDC are presented during training. Instead of pretraining on Cityscapes and then fine-tuning on ACDC (which risks catastrophic forgetting), this thesis uses an *alternate-batch* training strategy. Concretely, the training loop interleaves samples from the two datasets, so that every optimisation step is conditioned either on a clear-weather or on an adverse-weather image. This schedule ensures that the model continuously sees both domains throughout training, instead of first saturating on one and then being overwritten by the other. It also makes it straightforward to apply different optimisation behaviour depending on the current domain, which leads to the third design choice.

3) Selective encoder freezing to preserve clear-weather performance. The Mask2Former backbone (Swin-L) acts as a feature extractor for both domains. Since it is pretrained on Cityscapes, it already contains a strong representation for clear daytime scenes. The challenge is to adapt this representation to adverse conditions without destroying what it has already learned.

To this end, the thesis implements *selective encoder freezing*:

- For **Cityscapes batches**, the encoder (backbone) parameters are frozen: gradients with respect to the encoder weights are set to zero. Only the segmentation decoder and auxiliary heads are updated.
- For **ACDC batches**, the encoder is fully trainable. Gradients flow through the entire network, allowing the backbone to adapt its features to the appearance changes introduced by rain, fog, snow and night.

In effect, the model spends most of its adaptation capacity on the adverse domain, while keeping the clear-weather representation stable. Cityscapes acts as the strong, fixed anchor, and ACDC is the domain that needs additional capacity. The exact training algorithm and encoder update rules are detailed in Chapter 4.

4) Weather- and time-aware auxiliary supervision (WAS/TAS). The fourth choice is to exploit the condition metadata available in ACDC in the form of weather and time-of-day labels. For ACDC training images, a three-way weather label (rain, fog, snow) and a two-way time label (day, night) are derived from the dataset subsets. Cityscapes images are implicitly clear/day but are not associated with explicit WAS/TAS targets; they contribute only via the segmentation loss.

This information is used to define two lightweight auxiliary heads:

- a *Weather-Aware Supervision (WAS)* head that predicts the adverse-weather type (rain, fog, snow) from the encoder features;
- a *Time-Aware Supervision (TAS)* head that predicts day vs night.

During training, their classification losses are added to the main segmentation loss with very small weights, so that they act as gentle regularisers without dominating the optimisation. At

inference time, the WAS and TAS heads are removed entirely; the deployed model is identical, in structure and runtime cost, to a standard Mask2Former network. This follows the general idea that carefully chosen auxiliary tasks can regularise and improve the main task in road-scene understanding [13]. Architectural details and loss definitions are given in Chapter 4.

Summary of the approach. In summary, the thesis tackles the “one model for all conditions” goal by: (i) starting from a strong transformer-based baseline pretrained on clear-weather Cityscapes (Mask2Former + Swin-L); (ii) exposing the model to both Cityscapes and ACDC throughout training via alternate-batch sampling; (iii) preserving clear-weather performance through selective encoder freezing on Cityscapes batches while allowing adaptation on ACDC batches; and (iv) gently regularising the encoder with weather- and time-aware auxiliary supervision that is active only at training time on ACDC images. The effectiveness of these design choices is evaluated quantitatively and via ablation studies in the later chapters.

1.5 Research Questions

The overarching objective of this thesis is to design, implement and evaluate a transformer-based semantic segmentation framework that is robust across both clear and adverse conditions, using only real-world datasets. Concretely, the work focuses on adapting Mask2Former with a Swin-L backbone [4], [5], pretrained on Cityscapes [2], to also perform well on the ACDC dataset, which covers rain, fog, snow and night scenes [3].

From this objective, the following research questions are formulated:

- **RQ1:** How does a Cityscapes-pretrained Mask2Former model perform on adverse-condition images from ACDC without any adaptation?
- **RQ2:** Can alternate-batch training on Cityscapes and ACDC, combined with selective encoder freezing, improve performance on ACDC while preserving Cityscapes performance?
- **RQ3:** Does adding auxiliary Weather-Aware Supervision (WAS) and Time-Aware Supervision (TAS) heads during training, supervised on ACDC condition labels, further improve robustness, particularly in the most challenging conditions such as night-time?
- **RQ4:** How does the proposed transformer-based approach compare against a DeepLabV3+ baseline under the same training regimes (Cityscapes only, Cityscapes+ACDC and Cityscapes+ACDC with WAS/TAS)?

Answering these questions requires a systematic comparison across models, training strategies and evaluation conditions. The thesis therefore defines a consistent experimental protocol for both Cityscapes and ACDC, including per-weather evaluation on ACDC and ablation studies that isolate the contributions of individual components [2], [3], [7].

1.6 Contributions

This thesis makes the following main contributions:

- **C1: Transformer-based adverse-condition segmentation pipeline.** A semantic segmentation pipeline is built on Mask2Former with a Swin-L backbone [4], [5], starting from a CS-pretrained checkpoint and extending it to clear- and adverse-weather conditions on CS and ACDC [2], [3].
- **C2: Alternate-batch training with selective encoder freezing.** An alternate-batch strategy interleaves CS and ACDC mini-batches: on CS batches the encoder is frozen to preserve clear-weather features, while on ACDC batches the full model is updated to adapt to adverse conditions.
- **C3: Weather- and time-aware auxiliary supervision (WAS/TAS).** Two lightweight heads predict weather (rain, fog, snow) and time of day (day, night) for ACDC images. They are used only during training, with small loss weights, to encourage weather- and illumination-aware features without changing the inference architecture [7], [13].
- **C4: Comprehensive evaluation and ablation on CS and ACDC.** The method is evaluated on the official validation splits of CS and ACDC using the 19-class label space [2], [3]. Comparisons with DeepLabV3+ and Mask2Former baselines under multiple training regimes, together with a staged ablation, quantify the impact of alternate-batch training, encoder freezing and WAS/TAS on overall and per-condition mIoU [1], [4], [7], [10], [11].
- **C5: Demonstration of improved robustness without sacrificing clear-weather performance.** The final model achieves approximately 0.83 mIoU on CS and 0.77 on ACDC, corresponding to an ACDC gain of about 14 percentage points over the CS-only Mask2Former baseline, while maintaining strong clear-weather performance.

1.7 Thesis Outline

This thesis is organised as follows. Chapter 2 reviews background and related work on semantic segmentation, transformer-based architectures and robustness under adverse conditions. Chapter 3 introduces the CS and ACDC datasets, describes the train/validation/evaluation splits and formalises the task of learning a single model that is robust across weather and time-of-day conditions. Chapter 4 presents the proposed Mask2Former-based method, including the WAS/TAS heads, the alternate-batch training scheme and selective encoder freezing. Chapter 5 details the experimental setup (implementation, hardware, hyperparameters and evaluation protocol). Chapter 6 reports and analyses the quantitative and qualitative results, including baseline comparisons and ablation studies. Finally, Chapter 7 summarises the main findings, discusses limitations and outlines directions for future work, such as extending the approach to synthetic adverse-weather datasets and to anomaly detection under adverse conditions.

2 Background and Related Work

2.1 Semantic Segmentation in Autonomous Driving

Semantic segmentation assigns a semantic label to every pixel in an image, producing a dense representation of the scene that includes roads, vehicles, pedestrians, buildings, vegetation, and other categories relevant to autonomous driving [1], [6]. Compared to object detection or instance segmentation, semantic segmentation captures fine spatial detail and continuous regions, which is useful for identifying drivable areas, static obstacles, sidewalks, and scene boundaries [14]. This makes it a core component of the perception stack in modern driver assistance and autonomous driving systems.

Widely used semantic segmentation datasets in autonomous driving—including Cityscapes [2], BDD100K [15], and Mapillary Vistas [16]—contain mainly clear, daytime images collected in relatively controlled environments. Models trained on these datasets achieve high accuracy on in-distribution evaluations but often fail to generalise when deployed in different conditions such as rain, fog, snow, low illumination, or night-time scenes [3], [7]. This domain gap between training and deployment conditions is one of the key challenges addressed in this thesis.

2.2 CNN-Based Architectures

Early semantic segmentation architectures were built on convolutional neural networks (CNNs) designed originally for image classification. Fully Convolutional Networks (FCN) introduced dense prediction by replacing fully connected layers with convolutional layers and using upsampling to recover resolution at the output [6]. U-Net extended this idea with a symmetric encoder–decoder structure and skip connections, improving boundary accuracy and performance on limited data [17].

DeepLabV3+ became one of the most widely used architectures due to its atrous spatial pyramid pooling (ASPP) and encoder–decoder design, which allow multi-scale context aggregation while maintaining reasonable spatial resolution [1]. It supports modern backbones such as Xception, ResNet, and MobileNet and achieves strong results on challenging benchmarks like Cityscapes [2].

Despite these successes, purely CNN-based models have limitations when scenes become ambiguous, textures degrade, or long-range dependencies are important. Under adverse weather conditions, CNNs often struggle with reduced visibility, glare, and complex lighting effects, which motivates the adoption of transformer-based architectures that model global context more explicitly.

2.3 Transformers for Semantic Segmentation

Transformers were first introduced for sequence modelling in natural language processing, where self-attention replaces recurrent connections and convolutions [18]. Vision Transformer (ViT) showed that a pure transformer architecture can also be applied to images by splitting them into patches and treating them as tokens [19]. Subsequent work extended transformers to dense prediction and segmentation tasks [20].

Mask2Former applies the transformer paradigm to segmentation by formulating it as a mask-classification problem [4]. Instead of predicting pixel labels independently, Mask2Former uses a fixed set of learnable queries, each responsible for a region or object, to produce binary masks and associated class labels. A pixel decoder aggregates multi-scale feature maps from the backbone and feeds them into the transformer decoder, improving instance separation and robustness to clutter [4].

When combined with hierarchical backbones such as Swin Transformer, which applies shifted windows to efficiently model local and global context [5], Mask2Former achieves state-of-the-art performance on datasets like Cityscapes and COCO [4]. These results motivate the use of Mask2Former with a Swin backbone as the base architecture in this thesis.

2.4 Semantic Segmentation Under Adverse Conditions

Adverse weather poses several challenges for vision-based perception:

- **Visibility degradation:** fog, snow, and heavy rain introduce haze, scattering, and streaks, which reduce contrast and blur object boundaries [8], [21], [22].
- **Illumination variation:** night-time scenes and low-light conditions reduce dynamic range and increase sensor noise, making small objects and distant structures harder to detect [3], [9].
- **Texture ambiguity:** wet roads, snow-covered surfaces, and reflections from puddles can cause confusion between road, sidewalk, and background classes [7].
- **Domain shift:** models trained on clear-weather images do not naturally generalise to adverse-weather conditions without explicit adaptation [3], [7].

The ACDC dataset was introduced to benchmark robustness under four adverse conditions: rain, snow, fog, and night [3]. It provides pairs of corresponding images between clear and adverse conditions for the same locations, enabling detailed analysis of domain shift. However, manual annotation of such data is significantly more costly than for standard clear-weather datasets, which limits the availability of labelled adverse-weather data and motivates methods that use existing clear-weather datasets more effectively [3].

Kerim et al. proposed a framework for semantic segmentation under adverse conditions that combines synthetic data generation, weather-aware normalisation, and domain adaptation [7]. Their approach relies on DeepLabV3+ as the backbone and uses an Adverse Weather Synthetic Set (AWSS) to augment scarce real adverse-weather images. In contrast, the work in this thesis focuses on a transformer-based backbone and uses only real images from Cityscapes and ACDC without synthetic augmentation.

2.5 Domain Adaptation and Multi-Domain Training

Domain adaptation techniques aim to reduce the performance drop between a source domain (typically clear weather) and a target domain (adverse weather) by aligning distributions or learning domain-invariant features. Approaches include adversarial domain adaptation [23], [24], self-ensembling and consistency regularisation [25], synthetic-to-real adaptation using rendered driving datasets [26], and specialised methods for foggy or rainy scenes [21], [22].

For semantic segmentation in autonomous driving, synthetic datasets and rendering pipelines have been used to generate large numbers of labelled images under adverse conditions, which are then adapted to real data [24], [26], [27]. However, synthetic images may introduce a gap in style, texture, or geometry compared to real data, and the adaptation process can be complex and computationally expensive [7].

Multi-domain training offers a different strategy when some labelled data is available in each domain. By mixing batches from clear- and adverse-weather datasets, the model can benefit from the richness of the source domain while gradually adapting to the target domain [7]. Naive joint training, however, can degrade performance on the source domain if the model overfits to the more challenging target domain or if the training dynamics are not properly balanced.

In this thesis, a multi-domain training strategy is adopted in which clear- and adverse-weather samples are interleaved during training, with explicit mechanisms to preserve clear-weather performance while improving robustness to adverse conditions. The concrete training scheme, including sampling schedules and encoder update rules, is described in detail in Chapter 4.

2.6 Auxiliary Supervision: Weather and Time Awareness

Auxiliary supervision is a form of multi-task learning in which additional prediction tasks are introduced during training to encourage more informative feature representations [13], [28], [29]. In semantic segmentation, auxiliary tasks such as depth estimation, surface normals, or scene attributes can help the network learn richer geometric and contextual information that benefits the main segmentation task [13], [22].

For adverse-weather perception, scene-level attributes such as weather and time of day are particularly relevant. They capture global characteristics—for example, low visibility, strong

glare, or night-time illumination—that strongly influence local appearance and can guide the network towards more robust features.

Motivated by this, the thesis uses auxiliary tasks based on weather and time-of-day information derived from the ACDC dataset. In particular, Weather-Aware Supervision (WAS) and Time-Aware Supervision (TAS) heads are introduced that operate on the encoder features and are trained jointly with the segmentation objective. For ACDC images, WAS predicts the adverse-weather category (rain, fog, snow), while TAS predicts the time of day (day vs. night). Cityscapes images contribute only through the segmentation loss and are not associated with explicit WAS/TAS targets.

The losses of these auxiliary heads are weighted by small coefficients so that they guide feature learning without dominating the optimisation. The detailed architecture of these heads and their integration into the training objective are presented in Chapter 4.

This design differs from prior work that focuses mainly on synthetic augmentation and domain adaptation for robustness under adverse conditions [7], [8], [9], [21]. Instead, the goal is to exploit real-world condition metadata from ACDC to regularise a transformer-based segmentation model trained jointly on Cityscapes and ACDC.

2.7 Summary

This chapter reviewed the evolution of semantic segmentation models, the challenges posed by adverse weather, and the limitations of current models when generalising across domains. CNN-based architectures such as FCN, U-Net, and DeepLabV3+ laid the foundation for dense prediction, while transformer-based models such as Mask2Former with Swin backbones now provide state-of-the-art performance on standard benchmarks [1], [4], [5], [6], [17]. However, both classes of models still suffer from performance drops when moving from clear to adverse weather [3], [7].

Prior work has explored domain adaptation, synthetic data generation, multi-domain training, and auxiliary tasks to improve robustness [7], [21], [23], [24], [26]. Building on these ideas, this thesis proposes a transformer-based framework that combines multi-domain training on Cityscapes and ACDC, explicit mechanisms to preserve clear-weather performance, and weather- and time-aware auxiliary supervision on ACDC. The next chapter describes the datasets used and formalises the problem addressed in this thesis.

3 Datasets

3.1 Overview

This chapter describes the datasets used in this thesis, the characteristics of each weather and illumination domain, the annotation format, and the 19-class semantic label space. Because the proposed method aims to train a *single model* that performs robustly under both clear and adverse conditions, it is important to understand the distribution and structure of the datasets used: Cityscapes (clear-weather domain) and ACDC (adverse-weather domain) [2], [3].

Both datasets provide high-quality pixel-level annotations for urban driving scenes and share the same 19 semantic classes. Cityscapes focuses on clear daytime images from European cities, while ACDC focuses on adverse-weather and night-time scenes in similar environments. This complementary design makes them a natural pair for investigating cross-weather robustness without changing the label space.

Crucially, **both Cityscapes and ACDC consist entirely of real-world images**; no synthetic or simulated data are used anywhere in this thesis. This contrasts with approaches that rely on synthetic datasets such as SHIFT [26] or GTA-style renderings, and keeps all evaluations anchored in real driving conditions.

Another important difference is that ACDC is explicitly organised into four adverse condition domains (rain, fog, snow, night) and provides per-image metadata indicating the corresponding subset [3]. Cityscapes, in contrast, does not provide explicit weather or time-of-day labels and consists only of clear daytime scenes [2]; in this thesis all Cityscapes images are therefore treated as belonging to a single “clear/day” domain. For evaluation, we use the official validation splits of Cityscapes (500 images) and ACDC (406 images) as held-out evaluation sets; the Cityscapes official test set is not used.

3.2 Cityscapes Dataset

Cityscapes is a large-scale, real-world dataset collected across 50 European cities under clear, daytime weather conditions [2]. It was originally designed for evaluating semantic segmentation in normal driving conditions and has become one of the most widely used benchmarks in autonomous driving research.

In this thesis, Cityscapes serves as the clear-weather reference domain and provides strong supervision for common urban semantics. Only the *fine* pixel-level annotations are used to ensure consistent label quality and comparability with ACDC [2].

3.2.1 Image Characteristics

Cityscapes images are captured using a forward-facing stereo camera mounted on a vehicle driving in urban environments. The key properties are:

- **Resolution (raw):** 2048×1024 pixels (high-resolution RGB images),
- **Environment:** urban European cities with roads, buildings, traffic infrastructure, and pedestrians,
- **Lighting:** daytime only (no dawn, dusk, or night scenes),
- **Weather:** predominantly clear, with only mild variations in cloud cover,
- **Condition labels:** no explicit per-image weather or time-of-day annotation is provided; all images are implicitly clear/day.

During training in this thesis, the raw images are resized and cropped to a fixed input resolution suitable for the available GPU; the exact preprocessing pipeline is described in Chapter 5.

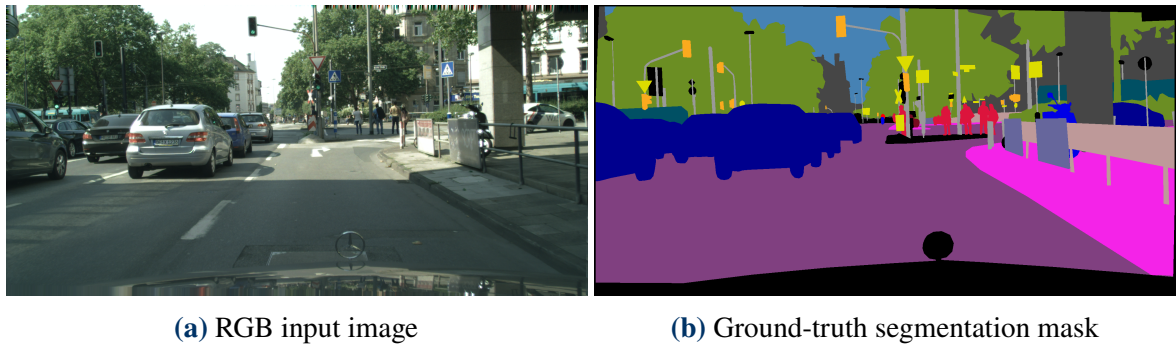


Figure 3.1: Example Cityscapes image and corresponding 19-class semantic ground-truth mask from the clear/day domain [2]. Each colour encodes one of the Cityscapes classes (road, sidewalk, building, vehicle types, etc.).

3.2.2 Annotation Format

Cityscapes provides dense pixel-level annotations with an official 19-class label space that has become a standard in urban scene understanding [2]. The annotations include typical traffic-related categories such as roads, sidewalks, buildings, walls, fences, poles, traffic lights, traffic signs, vegetation, terrain, sky, persons, riders, and different vehicle types (car, truck, bus, train, motorcycle, bicycle). Each semantic class is represented by a unique colour in the ground-truth mask, as illustrated in Figure 3.1. Annotations are created manually by expert annotators and have high geometric and semantic accuracy, which is important for training and evaluating segmentation models.

3.2.3 Dataset Splits

The official Cityscapes dataset provides:

- **Train (official):** 2975 finely annotated images,
- **Validation (official):** 500 finely annotated images,
- **Test (official):** 1525 images (labels not publicly available, used for leaderboard submissions).

Split terminology used in this thesis. The official **train** split is divided into an internal **train/val** split used for checkpoint selection:

- **Train (internal, 85%):** 2529 images,
- **Val (internal, 15%):** 446 images.

All final Cityscapes results reported in Chapter 6 are computed on the **official validation split** (500 images), which is treated as the held-out evaluation set for this thesis. The Cityscapes official test set is not used.

3.3 ACDC Dataset

The Adverse Conditions Dataset with Correspondences (ACDC) is a real-world dataset designed specifically to evaluate segmentation robustness under challenging weather and illumination conditions [3]. Unlike many synthetic or simulation-based datasets, ACDC consists entirely of real outdoor images collected in European cities under adverse conditions.

3.3.1 Weather and Illumination Domains

ACDC is organised into four subsets corresponding to different types of adverse conditions:

- **Rain:** rainy scenes with wet roads, reflections, and raindrop streaks,
- **Fog:** foggy scenes with strong atmospheric scattering and reduced visibility,
- **Snow:** snow-covered roads and sidewalks, reduced contrast, and occluded road markings,
- **Night:** low-light and night-time scenes with artificial illumination and strong contrast variations.

Each image comes with a label indicating which of these four domains it belongs to, providing explicit per-image weather or illumination metadata that can be used for condition-aware supervision. Many locations in ACDC have corresponding images captured in clear and adverse conditions, which allows detailed analysis of how weather affects segmentation performance [3].

3.3.2 Image Characteristics

The main properties of ACDC images are:

- **Resolution (raw):** 1920×1080 pixels (resized and cropped during training),
- **Geographic variation:** scenes from multiple cities and locations, overlapping with Cityscapes regions,
- **Visual disturbances:** haze, glare, reflections, motion blur, sensor noise, and occlusions due to snow and rain.

These conditions create strong domain shift relative to Cityscapes, particularly in low-contrast regions, distant structures, and small objects. As with Cityscapes, the training pipeline in this thesis resizes and crops the images to a fixed working resolution; implementation details are given in Chapter 5.

3.3.3 Annotation Format

Like Cityscapes, ACDC provides high-quality dense semantic labels using the same 19-class label space [3]. The annotation format (class IDs, colour scheme, and label meaning) is consistent with Cityscapes, as illustrated in Figure 3.2. This enables:

- direct training on the union of Cityscapes and ACDC,
- cross-dataset evaluation without label remapping,
- per-weather analysis of the same semantic categories.

This compatibility is a key reason for choosing ACDC as the adverse-weather dataset in this thesis.

3.3.4 Dataset Splits

The official ACDC splits are [3]:

- **Train (official):** 1600 images,
- **Validation (official):** 406 images,
- **Test (official):** additional images without public labels.

Split terminology used in this thesis. The official **train** split is divided into an internal **train/val** split used for checkpoint selection:

- **Train (internal, 85%):** 1360 images,
- **Val (internal, 15%):** 240 images.



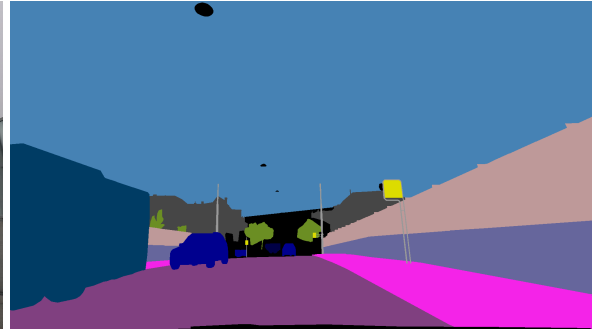
(a) Rain – RGB input



(b) Rain – ground truth



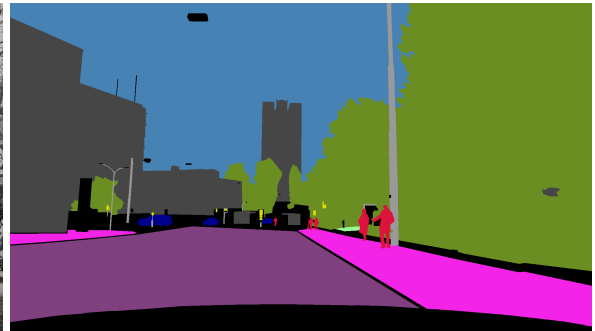
(c) Fog – RGB input



(d) Fog – ground truth



(e) Snow – RGB input



(f) Snow – ground truth



(g) Night – RGB input



(h) Night – ground truth

Figure 3.2: Example ACDC images and corresponding 19-class ground-truth masks for the four adverse conditions: rain, fog, snow, and night [3]. The ground-truth masks use the same colour encoding and label definitions as Cityscapes, which allows direct cross-weather comparison.

All final ACDC results reported in Chapter 6 are computed on the **official validation split** (406 images), which is treated as the held-out evaluation set for this thesis.

The ACDC evaluation set used in this thesis contains:

- 100 rain images,
- 100 fog images,
- 100 snow images,
- 106 night images.

These per-condition splits are used in Chapter 6 to report separate performance for rain, fog, snow, and night.

3.4 Cityscapes vs. ACDC: Dataset Comparison

Table 3.1 summarises the high-level properties of the two datasets, while Table 3.2 lists the exact train/val/evaluation splits used in this thesis.

Property	Cityscapes	ACDC
Resolution (raw)	2048×1024	1920×1080
Domain	clear/daytime	rain, fog, snow, night
Scenes	European cities	similar European cities
Label space	19 classes	19 classes (Cityscapes-compatible)
Image source	real	real
Explicit weather/illumination labels	no	yes (rain/fog/snow/night)
Main purpose	normal-weather segmentation	adverse-weather robustness

Table 3.1: High-level properties of the Cityscapes and ACDC datasets [2], [3].

Dataset	Domain	Train (85%)	Val (15%)	Official val (evaluation)
Cityscapes	clear daytime	2529	446	500
ACDC	rain+fog+snow+night (all)	1360	240	406

Table 3.2: Dataset splits used in this thesis. The official training splits are divided into an 85% training and 15% validation split using a fixed seed. The official validation splits are used for final evaluation (not used during training). Both Cityscapes and ACDC contain only **real-world** images [2], [3].

For ACDC, the 406-image evaluation set is further divided by weather condition (100 rain, 100 fog, 100 snow, 106 night), which is used to report per-weather performance in Chapter 6. Cityscapes does not provide such per-condition splits and is treated as a single clear/day domain.

3.5 19-Class Semantic Label Space

Both datasets use the standard 19 Cityscapes classes with train IDs from 0 to 18 [2], [3]. Table 3.3 lists all classes together with their train ID and the corresponding colour from the official Cityscapes palette. This palette is used consistently for both ground-truth masks and visualisations throughout the thesis.

ID	Class	Colour (Cityscapes palette)
0	road	RGB (128, 64, 128)
1	sidewalk	RGB (244, 35, 232)
2	building	RGB (70, 70, 70)
3	wall	RGB (102, 102, 156)
4	fence	RGB (190, 153, 153)
5	pole	RGB (153, 153, 153)
6	traffic light	RGB (250, 170, 30)
7	traffic sign	RGB (220, 220, 0)
8	vegetation	RGB (107, 142, 35)
9	terrain	RGB (152, 251, 152)
10	sky	RGB (70, 130, 180)
11	person	RGB (220, 20, 60)
12	rider	RGB (255, 126, 162)
13	car	RGB (0, 0, 142)
14	truck	RGB (0, 0, 70)
15	bus	RGB (0, 60, 100)
16	train	RGB (0, 80, 100)
17	motorcycle	RGB (0, 0, 230)
18	bicycle	RGB (119, 11, 32)

Table 3.3: Cityscapes train IDs (0–18), class names, and corresponding colours in the official palette used for both Cityscapes and ACDC [2], [3].

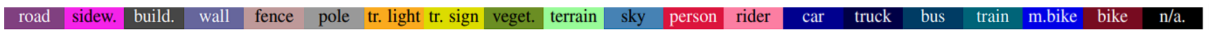


Figure 3.3: Cityscapes colour legend for the 19 semantic classes, adapted from Yang et al. [10]. The order and colours match the entries in Table 3.3.

Using the same label set and colour palette for both Cityscapes and ACDC ensures a consistent training objective and makes it straightforward to compare predictions across clear and adverse-weather conditions.

3.6 Summary

This chapter provided an overview of the Cityscapes and ACDC datasets used in this thesis. Cityscapes represents clear daytime urban driving conditions without explicit weather labels, serving as the benchmark for normal-weather semantic segmentation. In contrast, ACDC provides carefully curated real-world examples of four challenging adverse conditions—rain, fog, snow, and night—with explicit per-image condition annotations that enable condition-aware supervision.

Both datasets share the same 19-class semantic label space and annotation format, allowing direct joint training and evaluation without label remapping. Crucially, both consist exclusively of real-world images, ensuring that findings are grounded in realistic driving conditions. During training, the official training splits of Cityscapes and ACDC were each divided into an internal 85

The substantial domain gap between Cityscapes’ clear daytime scenes and ACDC’s adverse conditions, combined with the limited availability of annotated adverse-weather data, creates a challenging but realistic testbed for developing robust segmentation models. The next chapter presents the proposed method, which leverages this dataset foundation through a transformer-based architecture with weather- and time-aware supervision, selective encoder freezing, and alternate-batch training to address these challenges and achieve the goal of a single model performing well across all conditions.

4 Proposed Method

4.1 Overview

The goal of this work is to obtain a single semantic segmentation model that performs robustly under both clear-weather and adverse-weather conditions (rain, fog, snow, night) using only real-world data. Instead of training separate networks for each domain or fine-tuning a clear-weather model on adverse conditions (which typically hurts performance on the original domain), the method aims to keep strong performance on Cityscapes-style clear scenes while significantly improving on ACDC-style adverse scenes.

To reach this goal, four design decisions are combined into a single framework:

1. **Strong baseline architecture:** start from a state-of-the-art transformer-based segmentation model, **Mask2Former with a Swin-L backbone**, pretrained on Cityscapes.
2. **Alternate-batch training:** mix Cityscapes and ACDC batches during training so that the model is continuously exposed to both domains.
3. **Selective encoder freezing:** preserve clear-weather representations by freezing the encoder on Cityscapes batches and allowing full adaptation only on ACDC batches.
4. **Weather- and Time-Aware Supervision (WAS/TAS):** attach auxiliary weather and day/night classification heads to the encoder to guide the feature representation towards weather- and illumination-aware features on ACDC images.

In combination, these components form the proposed **Mask2Former-based architecture with Weather- and Time-Aware Supervision (WAS/TAS)**. Throughout this thesis, the WAS/TAS heads are supervised only on ACDC images, for which per-condition metadata is available; Cityscapes images contribute only through the segmentation loss and act as a clear/day reference domain. The exact pretrained checkpoint identifier and software environment used in the experiments are reported in Chapter 5.

Before converging on this design, several encoder–decoder architectures were considered, including DeepLabV3+ and SegFormer. These models are widely used and relatively simple to implement, but recent Cityscapes benchmarks show that transformer-based architectures such as Mask2Former clearly outperform classical encoder–decoder designs on urban street scenes [4], [12]. Given that the target application is also urban driving scenes, the method moves away from the original DeepLabV3+ baseline and builds directly on Mask2Former as the core model.

In practice, the implementation starts from a Mask2Former model pretrained on Cityscapes for 19-class semantic segmentation [4]. This backbone already captures rich clear-weather

structure, so training can focus on adapting it to adverse conditions rather than learning all features from scratch.

Component	Purpose
Mask2Former + Swin-L	Strong baseline for clear-weather segmentation
Alternate-batch training	Mix clear and adverse images during training
Selective encoder freezing	Preserve clear-weather features on Cityscapes
WAS/TAS heads	Encourage weather- and day/night-aware features (ACDC)

Table 4.1: Summary of the main components in the proposed method.

4.2 Original Mask2Former Architecture

Mask2Former is a universal image segmentation architecture that treats segmentation as a *mask-classification* problem, i.e. it predicts a fixed set of masks and their semantic labels instead of directly predicting a class for each pixel [4]. The same architecture can be used for panoptic, instance and semantic segmentation by simply changing the training targets. In this thesis, it is used purely as a semantic segmentation model, but the underlying structure follows the original work.

At a high level, the architecture consists of three main components:

1. **Backbone (Swin-L Transformer):** a hierarchical vision transformer that extracts multi-scale features from the input image [5];
2. **Pixel decoder:** a multi-scale feature aggregation module based on deformable attention that produces a high-resolution per-pixel feature map [4];
3. **Transformer decoder:** a set-based decoder that operates on a fixed number of learnable queries and predicts a set of mask embeddings and class logits [4].

Compared to earlier architectures that use per-pixel classification or bounding-box-based masks, three aspects of Mask2Former are particularly relevant for this work:

- **Masked attention:** instead of global cross-attention, Mask2Former restricts cross-attention to the predicted mask region of each query. This accelerates convergence and pushes attention towards object- or region-centric features rather than diffuse global context [4].
- **High-resolution feature pyramid:** the pixel decoder fuses features at multiple resolutions and produces a dense feature map at quarter resolution, which is shared by all queries. This is crucial for preserving fine details such as thin poles and object boundaries.
- **Set prediction with learnable queries:** each query is a learnable embedding that represents one potential region or object. During training, a Hungarian matching step assigns predicted

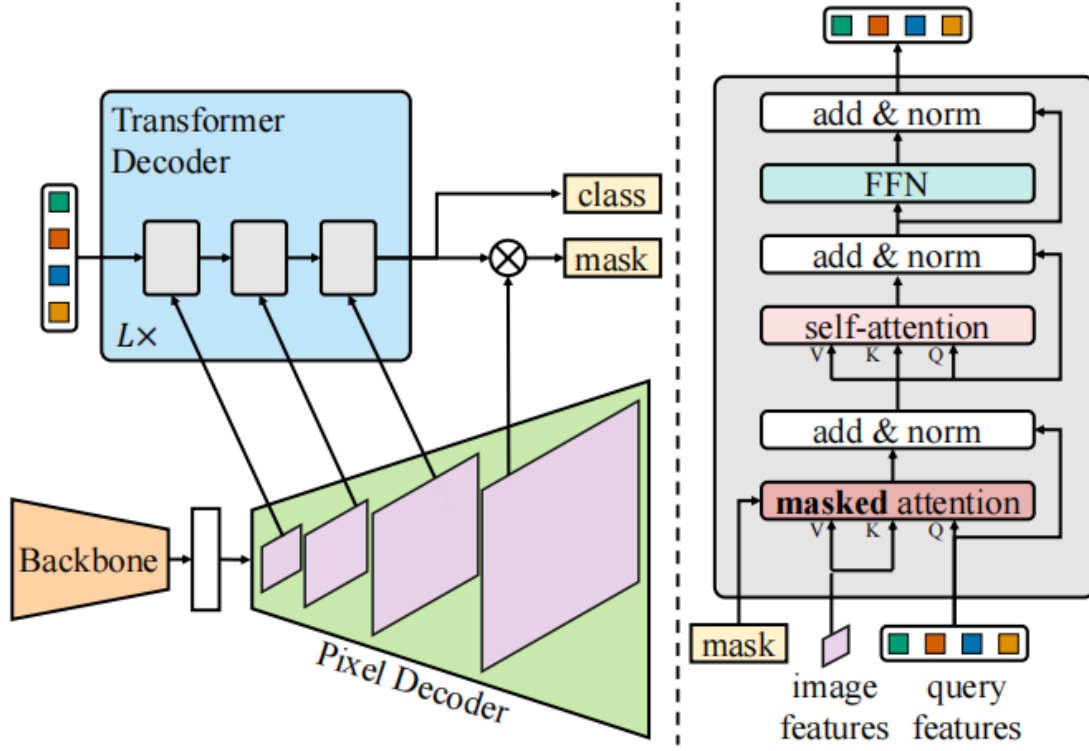


Figure 4.1: Original Mask2Former architecture with backbone, pixel decoder and masked-attention Transformer decoder, adapted from Cheng et al. [4].

masks to ground-truth segments, and the unmatched queries are trained to predict “no object”.

The following subsections briefly recap the three main blocks, focusing on the parts that are later re-used by the proposed method.

4.2.1 Backbone: Swin-L Transformer

The Swin-L backbone is a hierarchical Vision Transformer that computes self-attention within shifted windows [5]. In the training-phase explanation of Section 4.7, this component plays the role of a generic **feature extractor**: it receives the RGB image and outputs multi-scale feature maps that encode low-level edges and textures as well as high-level object regions.

The input image is first partitioned into non-overlapping patches, which are linearly projected into tokens and then processed by multiple stages. Each stage contains:

- **Window-based multi-head self-attention (W-MSA):** self-attention is computed within fixed-size windows, which keeps the computational cost linear in image size.
- **Shifted window mechanism (SW-MSA):** windows are shifted between blocks to allow information exchange across window boundaries.

- **Patch merging:** the spatial resolution is reduced while the channel dimension increases, building a feature pyramid.

This structure provides:

- strong long-range dependency modelling through repeated attention,
- high spatial resolution via the pyramid structure,
- excellent performance on dense prediction tasks, as demonstrated on Cityscapes and ADE20K [4], [5].

Choice of Swin-L variant. The Swin-L (Large) variant was selected over smaller alternatives (Swin-B, Swin-T) to maximize the model’s representational capacity and achieve the best possible segmentation accuracy. While smaller backbones offer computational advantages, our primary research objective was to establish the upper bound of performance for transformer-based adverse-weather segmentation and to clearly demonstrate the benefits over CNN-based alternatives like DeepLabV3+. Given the availability of sufficient computational resources (detailed in Chapter 5), we opted for the larger backbone to ensure that performance limitations were attributable to the proposed training strategy rather than model capacity.

In this thesis, the backbone is initialized from a Mask2Former model pretrained on Cityscapes [2], which gives a strong starting point for clear-weather scenes.

4.2.2 Pixel Decoder

The pixel decoder reconstructs high-resolution features from the backbone outputs. Mask2Former uses a pixel decoder based on multi-scale deformable attention [4], which aggregates information across resolutions and produces a dense feature map at one quarter of the input resolution. This map is shared by all queries in the Transformer decoder and is the basis for mask prediction. In intuitive terms, the pixel decoder **rebuilds a detailed per-pixel feature map** with sharper object boundaries and small-object structure.

4.2.3 Transformer Decoder with Masked Attention

The Transformer decoder operates on a fixed number of learnable query embeddings. At each decoder layer, it performs:

1. **Masked cross-attention:** each query attends only to the region inside its current predicted mask; the attention matrix outside the mask is suppressed.
2. **Self-attention:** queries exchange information with each other, which allows global context across regions.
3. **Feed-forward network:** a point-wise MLP that updates the query features.

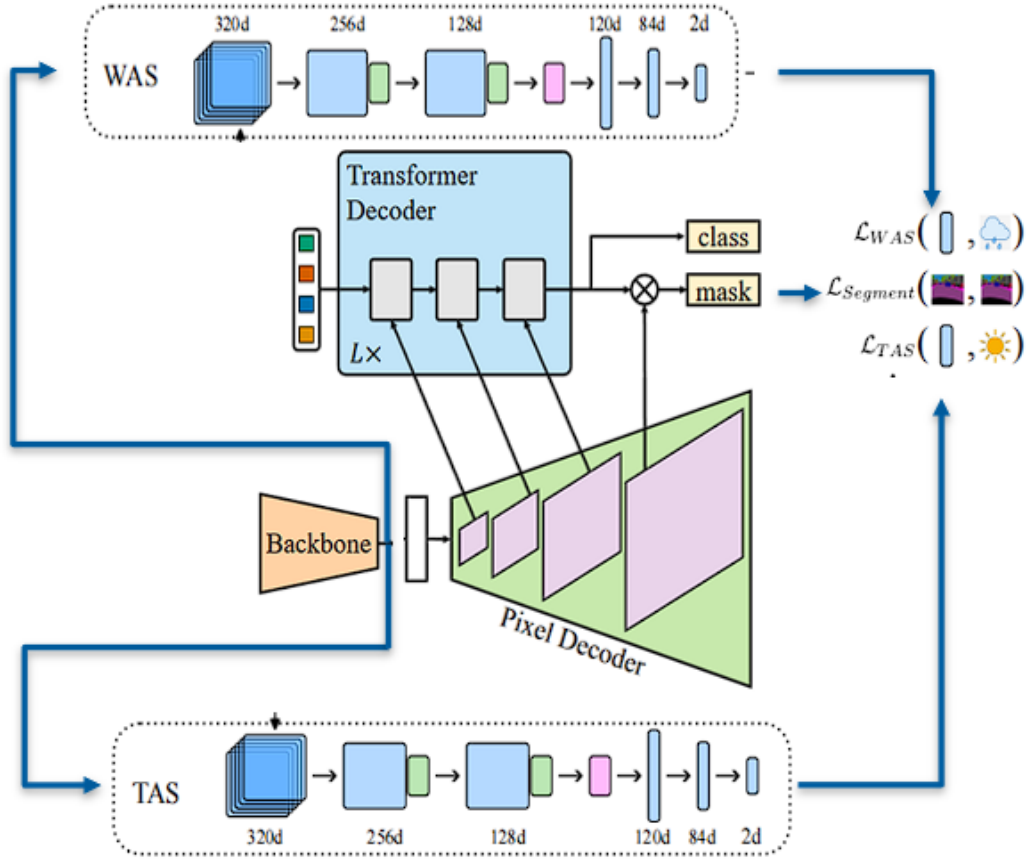


Figure 4.2: Proposed Mask2Former architecture with Weather-Aware Supervision (WAS) and Time-Aware Supervision (TAS). The Swin-L backbone feeds the pixel decoder and Transformer decoder (Mask2Former). In addition, global encoder features are passed to WAS and TAS heads during training, and segmentation and auxiliary losses are applied jointly.

After several decoder layers, each query produces a soft mask and a vector of class logits (including a “no object” class). A Hungarian matching step assigns predicted masks to ground-truth segments during training [4]. During inference, masks and classes are combined into a dense segmentation map by assigning each pixel to the highest-scoring active mask at that location.

4.3 Proposed Mask2Former Architecture with WAS/TAS

The proposed method augments the original Mask2Former with lightweight auxiliary classification heads for weather and day/night. These heads are attached to the encoder features and are active only during training when the corresponding labels are available. The core Mask2Former architecture and its segmentation head remain unchanged.

Relative to the original Mask2Former (Figure 4.1), the key differences are:

- two small MLP-based heads (WAS and TAS) are attached at the encoder output (or a selected intermediate feature map),
- the heads operate on globally pooled features, so they introduce only a negligible computational overhead during training,
- both heads are discarded for inference, so the deployed model has the same runtime cost and architecture as the base Mask2Former.

In the remainder of this chapter, this extended architecture is referred to as the **Mask2Former + WAS/TAS model**. The following sections explain how these auxiliary heads are used, how the joint loss is defined, and how the training strategy (alternate-batch schedule and encoder freezing) is applied on top of this architecture.

4.4 Weather- and Time-Aware Auxiliary Supervision

To improve robustness under adverse weather and night-time conditions, two auxiliary heads are added at the output of the backbone. This follows the general idea of multi-task learning, where auxiliary tasks are used to steer the shared encoder towards more informative representations [7], [28], [29], using the per-image condition metadata provided by ACDC. Cityscapes images do not provide explicit weather or day/night labels and therefore contribute only to the segmentation loss.

4.4.1 Weather-Aware Supervision (WAS)

The **Weather-Aware Supervision (WAS)** head predicts the adverse-weather category of the input image:

$$\text{Weather} \in \{\text{rain, fog, snow}\}.$$

Weather labels are derived from the ACDC condition subsets (rain, fog, snow). The WAS head consists of:

- global average pooling over the encoder feature map,
- a small stack of fully connected layers,
- a softmax classification layer.

WAS encourages the encoder to encode weather-sensitive cues such as haze patterns, snow textures, and raindrop streaks, instead of relying only on local semantic context. Because the segmentation head and the WAS head share the same encoder, any feature that helps to distinguish weather conditions is also available to the segmentation decoder.

4.4.2 Time-Aware Supervision (TAS)

The **Time-Aware Supervision (TAS)** head predicts illumination conditions:

$$\text{Time} \in \{\text{day}, \text{night}\}.$$

Night scenes introduce a strong domain shift compared to daytime due to:

- low overall illumination,
- strong shadows and highly localized light sources,
- reflections and glare from artificial lighting,
- increased sensor noise, especially in dark regions.

The TAS head helps the encoder to recognize and encode these illumination patterns. In this implementation, TAS supervision is applied only on ACDC images, where the night subset provides explicit night-time examples; Cityscapes images are implicitly day/clear but are not used to train TAS and instead contribute via the segmentation loss alone.

4.4.3 Training-Time Only

Both WAS and TAS are used only during training:

- the WAS/TAS heads and losses are active only when condition labels are available (i.e. on ACDC batches where rain/fog/snow/night metadata exists),
- on Cityscapes batches, where such metadata is not provided, training reduces to the standard segmentation objective without auxiliary losses,
- the WAS/TAS heads are discarded for inference, so the deployed model is identical to the original Mask2Former in structure and computational cost,
- at inference time, the model produces a single 19-class segmentation map, identical to the baseline output interface [4].

In other words, WAS and TAS act as a **pure training-time regulariser**: they encourage the encoder to learn weather- and day/night-aware features that improve robustness, without changing the architecture or runtime behaviour at deployment. This preserves fair comparability to the Mask2Former baseline and avoids any dependency on condition metadata at inference time.

4.5 Loss Functions

The total loss combines three components:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{seg}} + \alpha \mathcal{L}_{\text{WAS}} + \beta \mathcal{L}_{\text{TAS}}.$$

- $\mathcal{L}_{\text{seg}}$ is the segmentation loss used in Mask2Former, which combines classification and mask losses over the set of predicted masks [4].
- $\mathcal{L}_{\text{WAS}}$ is the cross-entropy loss for weather classification on ACDC images.
- $\mathcal{L}_{\text{TAS}}$ is the cross-entropy loss for day/night classification on ACDC images.

In this thesis, the auxiliary losses are deliberately down-weighted:

$$\alpha = \beta = 1 \times 10^{-5},$$

so that the WAS/TAS heads influence the encoder features without dominating the segmentation gradients. This specific weighting follows the practice established by Kerim et al. [7] for auxiliary tasks in adverse-weather segmentation, ensuring that the primary segmentation loss remains the dominant training signal while still providing gentle regularization through the auxiliary objectives. Empirically, this small weighting provides a modest but consistent improvement on adverse-weather scenes, especially at night, while leaving clear-weather performance essentially unchanged.

Gradient Flow and Parameter Updates

The three losses are computed separately and then summed into $\mathcal{L}_{\text{total}}$ for backpropagation. However, their gradients affect different parts of the network, following the idea of Kerim et al. [7] but adapted to the Mask2Former backbone:

- **Segmentation loss:** $\mathcal{L}_{\text{seg}}$ is backpropagated through the full segmentation branch, i.e. through the encoder (Swin-L), pixel decoder, Transformer decoder and the final segmentation head. It does not update the WAS/TAS heads.
- **WAS/TAS losses:** $\mathcal{L}_{\text{WAS}}$ and $\mathcal{L}_{\text{TAS}}$ are backpropagated only through the WAS/TAS heads and the encoder features that feed them. They do not update the pixel decoder or the Transformer decoder; their role is solely to shape the shared encoder representation towards weather- and day/night-aware features.
- **Cityscapes vs. ACDC batches:** on Cityscapes batches, weather/day-night labels are not available and therefore $\mathcal{L}_{\text{WAS}}$ and $\mathcal{L}_{\text{TAS}}$ are not evaluated; additionally, the encoder is frozen (Section 4.6.2), so encoder gradients are suppressed even for $\mathcal{L}_{\text{seg}}$. On ACDC batches, encoder-related gradients are allowed and all available losses are applied.

This matches the intended learning dynamics:

- $\mathcal{L}_{\text{seg}}$ drives the overall segmentation quality,
- $\mathcal{L}_{\text{WAS}}$ and $\mathcal{L}_{\text{TAS}}$ gently push the encoder to become weather- and day/night-aware (ACDC only),
- encoder freezing on Cityscapes preserves the strong clear-weather representation from the Cityscapes-pretrained Mask2Former model.

4.6 Training Strategy

While the choice of Mask2Former with a Swin-L backbone and WAS/TAS auxiliary heads is important, the central part of this work lies in *how* the model is trained on clear-weather and adverse-weather data. The proposed training strategy combines:

1. **alternate-batch multi-domain training**, and
2. **selective encoder freezing**.

Together, these mechanisms allow the network to maintain its strong performance on Cityscapes while significantly improving segmentation accuracy on ACDC across rain, fog, snow and night.

4.6.1 Alternate-Batch Multi-Domain Training

A central element of this work is the **alternate-batch training** strategy that mixes clear-weather and adverse-weather data at the level of individual gradient steps. Instead of pretraining on Cityscapes and then fine-tuning on ACDC (which typically causes catastrophic forgetting of clear-weather performance), batches from Cityscapes and ACDC are interleaved throughout training.

Conceptually, the training loop looks as follows:

- batch 1: images from Cityscapes,
- batch 2: images from ACDC,
- batch 3: images from Cityscapes,
- batch 4: images from ACDC,
- ...

For a batch size of one, this means individual images alternate between the two datasets. For larger batch sizes, each mini-batch still comes entirely from one dataset, but the *sequence of mini-batches* alternates between Cityscapes and ACDC.

This schedule ensures that:

- the model is continuously reminded of the clear-weather domain (stability on Cityscapes),
- the model gradually adapts to ACDC without drifting too far away from the clear-weather optimum,
- catastrophic forgetting of Cityscapes performance is reduced compared to a pure fine-tuning strategy on ACDC.

On top of this alternation, the loss is always computed over the standard 19-class segmentation objective, optionally combined with the WAS/TAS auxiliary losses described earlier in this chapter (ACDC only).

4.6.2 Selective Encoder Freezing

The Mask2Former + Swin-L baseline used in this work is already highly optimized for Cityscapes. If all encoder layers were freely updated on both datasets, the network would tend to drift towards ACDC and partially forget the clear-weather solution. To counter this, the encoder is frozen on Cityscapes batches and only updated on ACDC batches.

Throughout this thesis, the term *encoder* refers to the feature extraction backbone of Mask2Former (the Swin-L transformer), while *decoder* and *heads* refer to the pixel decoder, Transformer decoder and the segmentation/WAS/TAS prediction modules on top.

Cityscapes batches (clear-weather domain)

When the current batch comes from Cityscapes, the encoder is **frozen**:

- encoder parameters are kept fixed (no gradient updates),
- the segmentation decoder and segmentation head remain trainable,
- WAS/TAS supervision is not applied because Cityscapes does not provide weather/day-night labels.

Cityscapes batches are used to preserve and gently refine clear-weather segmentation performance without reshaping the core feature extractor.

ACDC batches (adverse-weather domain)

When the current batch comes from ACDC, the **entire model is trainable**:

- encoder parameters are updated,
- decoder and prediction heads are also updated,
- WAS/TAS losses are applied using ACDC condition metadata.

The model is allowed to adapt its features to adverse conditions (rain, fog, snow and night), while the decoder and heads learn to exploit these features for accurate pixel-wise predictions.

In practice, only a subset of encoder layers could be unfrozen to improve stability. In this thesis, the full encoder is updated on ACDC batches in our experiments.

Summary of batch behaviour

Batch source	Encoder	Main purpose
Cityscapes (clear)	frozen	preserve and slightly refine clear-weather performance
ACDC (adverse)	trainable	adapt features and predictions to adverse-weather conditions

Table 4.2: Summary of the proposed training strategy. Batches from Cityscapes and ACDC are processed in alternation. On Cityscapes batches, encoder parameters are frozen, while on ACDC batches the entire model is updated; WAS/TAS supervision is applied only on ACDC.

This combination of alternate-batch multi-domain training and selective encoder freezing is the key mechanism that enables the model to:

- maintain its performance on the clear-weather Cityscapes domain, and
- significantly improve segmentation accuracy on ACDC across rain, fog, snow and night.

Relation to previous work

The overall training strategy is inspired by the synthetic-aware approach of Kerim et al. [7], who introduced WAS/TAS-style auxiliary heads, multi-task learning, and an alternating schedule between clear-weather real data and adverse-weather *synthetic* data. However, this thesis differs in three key aspects:

- **Backbone:** instead of DeepLabV3+ with an Xception-based encoder, this work uses a transformer-based Mask2Former architecture with a Swin-L backbone.
- **Data:** only *real* datasets are used during training. CS plays the role of the standard clear-weather domain, and ACDC plays the role of the adverse-weather domain; no synthetic dataset is included.
- **Label space:** the full 19-class Cityscapes label space is preserved for both datasets, whereas Kerim et al. [7] work with a reduced 10-class mapping.

Conceptually, the encoder-freezing pattern is analogous: the encoder is frozen on clear-weather batches and updated on adverse-weather batches, while the decoder and prediction heads remain trainable. The novelty of this thesis is transferring this idea to a modern transformer-based segmentation architecture and validating it using only real-world data with the full 19-class label space.

4.7 Training Pipeline

During training, the full *Mask2Former* + WAS/TAS architecture (Figure 4.3) is active: the segmentation branch (backbone, pixel decoder, Transformer decoder) and the two auxiliary heads (WAS/TAS).

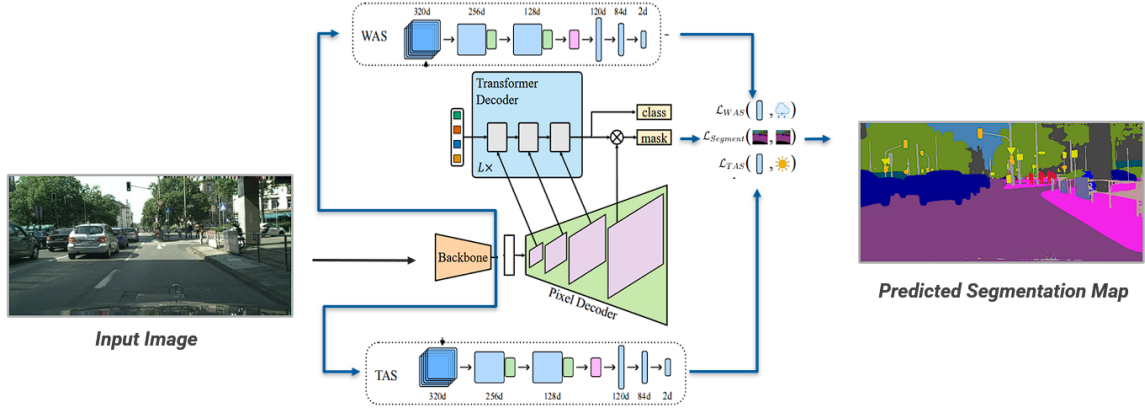


Figure 4.3: Training-phase pipeline of the proposed Mask2Former architecture with WAS/TAS. The network receives an input image and corresponding ground-truth labels. It predicts a segmentation map as well as weather and time classes when the auxiliary heads are active. The segmentation loss is always applied; WAS/TAS losses are used only when the corresponding labels exist.

Conceptually, each training step can be described by its inputs, outputs and applied losses.

Inputs. During training, the model operates on four types of ground-truth information:

- an RGB image,
- a semantic segmentation ground-truth mask,
- a weather label,
- a time-of-day label (day/night).

For datasets that do not provide weather or time-of-day annotations (e.g. Cityscapes), only the image and segmentation mask are available, and the auxiliary WAS/TAS losses are not evaluated.

Forward pass. The RGB image is processed by the Swin-L backbone, the pixel decoder and the Transformer decoder to produce a set of mask predictions and corresponding class scores. In parallel, the globally pooled encoder features are fed into the WAS and TAS heads to predict weather and time-of-day whenever these heads are enabled.

Outputs. The network produces up to three outputs:

- a predicted 19-class semantic segmentation map,

- a predicted weather class (rain, fog, snow),
- a predicted time-of-day class (day, night).

When weather or time labels are not available for a given sample, the corresponding auxiliary predictions and losses are ignored.

Losses and parameter updates. For every training batch, the segmentation loss $\mathcal{L}_{\text{seg}}$ is computed. When weather and time labels are present, the auxiliary losses $\mathcal{L}_{\text{WAS}}$ and $\mathcal{L}_{\text{TAS}}$ are added with small weights, as defined in Section 4.5. The encoder is frozen on Cityscapes batches and trainable on ACDC batches, in line with the strategy described in Section 4.6.2; decoder and heads used in training remain trainable in both cases. This perspective highlights that the key difference between batch types lies in which labels are available (segmentation vs. segmentation+weather+time) and whether the encoder parameters are updated.

4.8 Inference Pipeline

Inference refers to the *deployment* phase of the model, where the network is used to make predictions on previously unseen images without access to any ground-truth labels and without performing backpropagation. In this phase, the goal is to obtain a reliable 19-class segmentation map for each input image under real driving conditions.

During inference, the auxiliary WAS/TAS heads are removed and the model reduces to a standard Mask2Former with a Swin-L backbone. Only the segmentation branch is active; the weights have already been shaped during training by the combined effect of multi-domain learning and the auxiliary supervision.

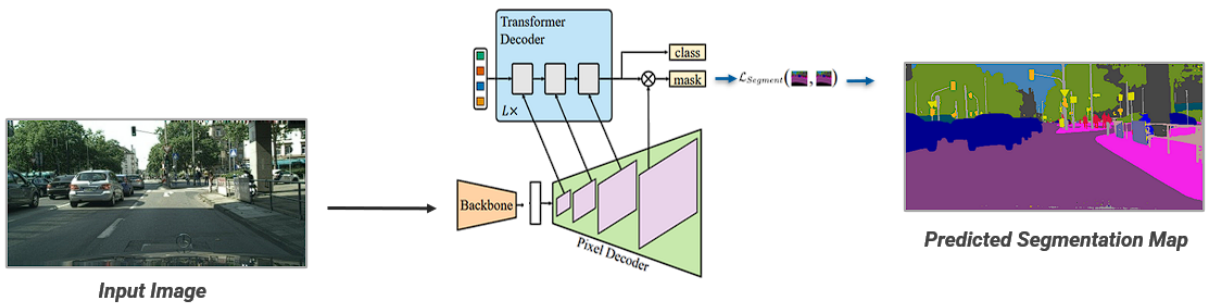


Figure 4.4: Inference-phase pipeline of the proposed Mask2Former architecture. Only the Mask2Former segmentation branch is kept. The network receives an RGB image and outputs a 19-class segmentation map; WAS/TAS heads are removed and do not incur any runtime cost.

The inference step can be described in the same input/forward-pass/output structure as the training pipeline, but without ground-truth supervision.

Inputs. At inference time, the model receives:

- a single RGB image from the on-board camera.

Forward pass. The image is processed through exactly the same segmentation branch as during training:

- the Swin-L backbone extracts multi-scale feature maps,
- the pixel decoder reconstructs a high-resolution feature map,
- the Transformer decoder predicts a set of masks and class scores.

The WAS/TAS heads are not present in this phase, so no auxiliary predictions are computed and no losses are evaluated.

Outputs. The set of predicted masks and classes is merged into a single:

- 19-class semantic segmentation map for the input image.

Each pixel is assigned to the class of the highest-scoring active mask at that location, following the standard Mask2Former inference procedure [4].

Because the WAS/TAS heads are removed, the deployed model has:

- no additional memory usage compared to the baseline Mask2Former,
- no runtime overhead,
- identical inference speed and complexity to the original Cityscapes-trained Mask2Former.

4.9 Summary

This chapter introduced the proposed transformer-based semantic segmentation framework for robust performance under clear and adverse weather. The method starts from a strong Mask2Former + Swin-L baseline and adds:

- alternate-batch training on Cityscapes and ACDC,
- selective encoder freezing to preserve clear-weather representations,
- lightweight weather- and day/night-aware auxiliary heads (WAS/TAS) for training-time supervision on ACDC images.

By combining multi-domain training with weak auxiliary supervision, the model is encouraged to learn features that remain effective under clear-weather conditions while adapting to challenging adverse-weather scenes without retraining separate networks. The next chapter details the experimental setup, including hyperparameters, training environment and evaluation protocols, followed by a quantitative and qualitative analysis of the resulting models.

5 Experimental Setup

5.1 Overview

This chapter describes the experimental setup used to train and evaluate the proposed semantic segmentation framework. It follows the structure commonly used in related work [7] and summarises the datasets, implementation details, baselines and evaluation metric before going into hardware, software and hyperparameters in more detail.

Datasets. All experiments are based on two real-world urban-scene datasets: Cityscapes and ACDC [2], [3]. For training, the official *training* split of each dataset is divided into an **85/15** split, which is used as **train/validation** during training. The 85/15 train/validation split is generated deterministically using a fixed holdout seed (`holdout_seed = 1`) to ensure the same samples are assigned to each split across runs.

For final evaluation, the official *validation* split of Cityscapes (500 images) and ACDC (406 images, spanning rain, fog, snow and night) is treated as the **held-out evaluation set**. Thus, during training we use an internal 85/15 split from the official training sets for training/validation, while final evaluation is performed on the official validation sets, which are treated as held-out evaluation sets for this thesis. Both datasets share the same 19-class label space and compatible colour-coding; detailed statistics and examples are given in the Datasets chapter (see Chapter 3).

Implementation details. All models are trained on a dedicated workstation described in Section 5.2, running Ubuntu 22.04.5 LTS with two NVIDIA RTX 4090 GPUs installed. **Only a single RTX 4090 GPU is used for all experiments;** the second GPU remains idle. DeepLabV3+ follows the configuration from Kerim et al. [7], while Mask2Former is implemented in PyTorch using a publicly available Mask2Former model with a Swin-L backbone initialised from Cityscapes semantic segmentation weights (`facebook/mask2former-swin-large-cityscapes-semantic`) and trained via a custom script built on top of the Hugging Face Transformers library [4], [5]. Within each model family, all three training variants (CS-only, CS+ACDC with alternate batches and encoder freezing, and the full WAS/TAS method) use exactly the same batch size, crop size and total iterations; only the training strategy changes. For Mask2Former, the total number of iterations is set to 90k, in line with the original Mask2Former paper on Cityscapes [4] and because the internal validation curves in this thesis converge around that point. For DeepLabV3+, 30k iterations are sufficient for convergence under all three training styles.

Baselines. Two backbone families are studied:

- **Baseline 1: DeepLabV3+** with a MobileNetV2 encoder (CNN-based baseline) [1], [7],
- **Baseline 2: Mask2Former** with a Swin-L backbone (transformer-based baseline) [4], [5].

For each backbone, three variants are trained: Cityscapes-only, Cityscapes+ACDC with alternate batches and encoder freezing, and the full proposed method with additional WAS/TAS auxiliary supervision. The variants are listed in detail in Chapter 6.

Evaluation metric. All quantitative comparisons use mean Intersection over Union (mIoU) over the common 19 classes on Cityscapes and ACDC, both overall and per-weather for ACDC; the precise definition and protocol are given in Section 5.6. Other metrics reported by the evaluation scripts (e.g. pixel accuracy) are not used for the main analysis.

The remainder of this chapter gives the hardware and software setup, dataset preprocessing, training configurations for DeepLabV3+ and Mask2Former, and the evaluation and checkpoint selection procedures.

5.2 Hardware Setup

All experiments in this thesis were conducted on a dedicated workstation running Ubuntu 22.04.5 LTS with an AMD Threadripper CPU and two NVIDIA RTX 4090 GPUs installed. **Only a single RTX 4090 GPU was used for all training and evaluation runs;** the second GPU remained unused during the experiments. The system provides sufficient compute and memory to train both DeepLabV3+ and Mask2Former with the crop sizes and batch sizes described in Sections 5.5.1 and 5.5.2.

5.2.1 Hardware Configuration

The main hardware specifications, derived from the workstation system diagnostics, are:

- **Operating System:** Ubuntu 22.04.5 LTS
- **Kernel:** Linux 6.8.0-85-generic
- **Processor:** AMD Ryzen Threadripper PRO 5975WX (32 cores, 64 threads)
- **System Memory:** 125 GiB DDR4 RAM
- **Graphics Processing Units (installed):** $2 \times$ NVIDIA GeForce RTX 4090 (24,564 MiB GDDR6X VRAM each)
- **GPUs used in experiments:** $1 \times$ NVIDIA GeForce RTX 4090
- **GPU Driver:** NVIDIA 535.274.02

Component	Specification
CPU	AMD Ryzen Threadripper PRO 5975WX (32C / 64T)
RAM	125 GiB DDR4
GPUs (installed)	2 × NVIDIA GeForce RTX 4090 (24,564 MiB each)
GPUs (used)	1 × NVIDIA GeForce RTX 4090
GPU driver	NVIDIA 535.274.02
Storage	3.6 TB NVMe SSD in /home (2.7 TB free)
OS	Ubuntu 22.04.5 LTS, kernel 6.8.0-85-generic

Table 5.1: Hardware configuration of the workstation used for all experiments. Although two RTX 4090 GPUs are installed, only one GPU was used in practice.

5.3 Software Stack

The DeepLabV3+ baseline is implemented using the PyTorch-based code from Kerim et al. [7], while Mask2Former is implemented using PyTorch with the Mask2Former + Swin-L model loaded from the Hugging Face Transformers library (`facebook/mask2former-swin-large-cityscapes-semantic`) and trained via a custom script.

All experiments were executed inside a dedicated conda environment (`mask2former-adverse`); the full dependency list is stored in `environment.yml` for reproducibility.

The main software components and versions used in the experiments are:

- **Python:** 3.8.20
- **PyTorch:** 2.0.0+cu118 (CUDA 11.8 runtime)
- **cuDNN:** 8700
- **Hugging Face Transformers:** 4.37.2
- CUDA-capable NVIDIA driver stack (see Section 5.2.1)
- standard Python libraries (NumPy, torchvision, OpenCV, etc.) for data loading and pre-processing

5.4 Dataset Preprocessing

Cityscapes and ACDC are preprocessed to ensure consistent training behaviour across domains [2], [3], [7]. Preprocessing steps follow standard practice in semantic segmentation, with a focus on real rather than synthetic adverse-weather conditions.

5.4.1 Cropping

During training, images are randomly cropped; the crop size differs between the two model families due to their memory footprint:

- **DeepLabV3+:** random crops of size 768×768 , consistent with the configuration in Kerim et al. [7],
- **Mask2Former:** random crops of size 640×640 , chosen to fit the Swin-L backbone and transformer decoder into GPU memory with a batch size of three images.

For each crop:

- the same spatial crop is applied to the RGB image and its ground-truth segmentation mask,
- crops are sampled fully inside the image (no padding is used),
- Cityscapes and ACDC follow the same procedure so that the model sees comparable field-of-view ranges in both domains.

5.4.2 Normalization

All images are normalised using ImageNet mean and standard deviation [30],

$$\mu = (0.485, 0.456, 0.406), \quad \sigma = (0.229, 0.224, 0.225),$$

which matches the pretraining regime used for Swin-L and ensures consistent input statistics for both DeepLabV3+ and Mask2Former.

5.4.3 Data Augmentation

Data augmentation is deliberately conservative to avoid introducing synthetic weather effects:

- random horizontal flipping,
- colour jitter,
- no artificial rain, fog, snow or night-time transformations.

This design mirrors the real-adverse-weather focus in Kerim et al. [7] and ensures that observed gains come from the training strategy and auxiliary supervision, not from synthetic domain expansion.

5.5 Training Configuration

This section summarises the training hyperparameters used for the DeepLabV3+ and Mask2Former experiments. For each architecture, a single configuration is used and then combined with the

different training strategies described in Chapter 4 (CS-only, CS+ACDC with alternate-batch + encoder freezing, and CS+ACDC with alternate-batch + encoder freezing + WAS/TAS).

5.5.1 DeepLabV3+ Training Setup

DeepLabV3+ is trained using the original implementation with an SGD-based optimiser and a polynomial learning-rate (poly) policy, following Kerim et al. [7]. The key hyperparameters used in this thesis are:

- **Number of classes:** 19,
- **Batch size:** 12,
- **Validation batch size:** 12,
- **Crop size:** 768×768 ,
- **Optimizer:** SGD,
- **Initial learning rate:** 0.3,
- **Learning-rate schedule:** poly policy with a step size of 10 000 iterations,
- **Total iterations:** 30 000,
- **Validation interval:** every 5 000 iterations,
- **Loss:** cross-entropy,
- **Weight decay:** 1×10^{-4} .

These hyperparameters are kept *identical* for all three DeepLabV3+ variants (CS-only, CS+ACDC with alternate-batch training and encoder freezing, and CS+ACDC with alternate-batch training, encoder freezing and WAS/TAS). Only the data scheduling and the presence or absence of encoder freezing and auxiliary heads are changed. This ensures that any performance differences between the three DeepLabV3+ runs are due to the training strategy and not to changes in batch size, crop size or total iterations.

5.5.2 Mask2Former Training Setup

Mask2Former with a Swin-L backbone is trained using AdamW, following the principles of the original Mask2Former configuration [4] and adapting them to the custom training script used in this thesis. The key hyperparameters for all Mask2Former experiments are:

- **Number of classes:** 19,
- **Batch size:** 3,

- **Validation batch size:** 3,
- **Crop size:** 640×640 ,
- **Optimizer:** AdamW,
- **Base learning rate:** 1×10^{-5} ,
- **Learning-rate schedule:** cosine schedule with warmup,
- **Total iterations:** 90 000,
- **Validation interval:** every 5 000 iterations,
- **Loss:** cross-entropy for the segmentation branch,
- **Weight decay:** 5×10^{-2} .

A smaller batch size and crop size are necessary here because Mask2Former with a Swin-L backbone is significantly heavier than DeepLabV3+ with a MobileNetV2 encoder. On the workstation described in Section 5.2, a batch size of 12 with 768×768 crops is not practical for Mask2Former, whereas a batch size of 3 with 640×640 crops provides a good compromise between memory usage and throughput.

It is also **not meaningful to force DeepLabV3+ and Mask2Former to share exactly the same training schedule**. Mask2Former is a transformer-based architecture with higher capacity and typically requires longer training to converge. In the original Mask2Former paper, Cityscapes models are trained for 90 000 iterations [4], and this thesis adopts the same iteration budget as a reference point.

Empirically, all Mask2Former runs were monitored until the *internal* validation mIoU curves (15% hold-out from the official training split) stabilised. These curves were logged live using Visdom (see Section 5.7.1), and 90 000 iterations corresponded to the regime where performance appeared to converge. For DeepLabV3+, validation curves stabilised earlier; 30 000 iterations were sufficient for convergence and further training did not bring meaningful improvements.

5.5.3 Summary of Core Hyperparameters

Table 5.2 summarises the core hyperparameters for both model families.

Model	Batch	Val batch	Crop size	LR	Total itrs	Val interval
DeepLabV3+	12	12	768×768	0.3	30 000	5 000
Mask2Former	3	3	640×640	1×10^{-5}	90 000	5 000

Table 5.2: Core training hyperparameters for DeepLabV3+ and Mask2Former. Within each model family, these values are kept identical across all training variants; only the training strategy (datasets, alternate batches, encoder freezing, WAS/TAS) changes.

For Mask2Former runs that include auxiliary tasks, the WAS and TAS loss terms are weighted by

$$\alpha = \beta = 1 \times 10^{-5}.$$

5.5.4 Alternate-Batch Pipeline and Encoder Freezing

For variants that use both Cityscapes and ACDC, a simple 1:1 alternate-batch schedule is used: one mini-batch from Cityscapes, then one from ACDC, and so on. This schedule is applied identically to both DeepLabV3+ and Mask2Former; the difference lies only in the backbone and optimiser.

For Mask2Former, **low-level encoder/backbone layers are frozen on Cityscapes batches** and the network is **fully trainable on ACDC batches**. This asymmetry allows the model to adapt its representation to adverse conditions while reducing drift of early clear-weather features learned from Cityscapes pretraining [7]. For DeepLabV3+, encoder freezing is implemented analogously in the MobileNetV2 backbone for the extended baselines.

5.6 Evaluation Protocol

5.6.1 Cityscapes Evaluation

Cityscapes evaluation is performed on the official validation split (500 images), which is treated as the **held-out evaluation set**, using the standard 19-class label space [2], [7]. All numbers reported in Chapter 6 follow this protocol.

5.6.2 ACDC Evaluation

ACDC evaluation is performed on the official validation split (406 images), which is treated as the **held-out evaluation set**.

Results are reported both:

- over the full **held-out evaluation set**, and
- separately for each subset: rain, fog, snow and night [3].

Label mappings follow the common 19-class scheme to ensure direct comparability with Cityscapes, as in Kerim et al. [7].

5.6.3 Metrics

The primary evaluation metric is mean Intersection over Union (mIoU) over the 19 classes. Pixel accuracy and mean class accuracy are also computed, but all main comparisons in Chapter 6 are based on mIoU.

5.7 Checkpoint Selection

Both DeepLabV3+ and Mask2Former training runs store periodic checkpoints and track the best-performing model on the **internal validation split** (15% hold-out from the official training split).

Two checkpoints are retained for each run:

- a **last** checkpoint containing the model state after the final iteration,
- a **best** checkpoint corresponding to the highest mIoU on the **internal Cityscapes validation split** during training.

We select the best checkpoint using Cityscapes internal validation to explicitly enforce preservation of clear-weather performance; ACDC internal validation is tracked in parallel to ensure robustness is not achieved by sacrificing Cityscapes.

For mixed-domain runs (CS+ACDC), internal validation is computed on both Cityscapes and ACDC at each validation interval, and the corresponding ACDC internal validation mIoU is recorded alongside the selected best checkpoint. All quantitative and qualitative results in Chapter 6 are reported on the **held-out evaluation sets** (official validation splits), using the selected best checkpoints.

5.7.1 Training Monitoring with Visdom

During training, the custom training script logs the segmentation loss and the internal validation mIoU to Visdom. Validation mIoU is tracked separately on Cityscapes and ACDC (using the 85/15 internal split) to monitor convergence and to motivate the 90 k iteration budget used for Mask2Former, as performance stabilises in the later stage of training.

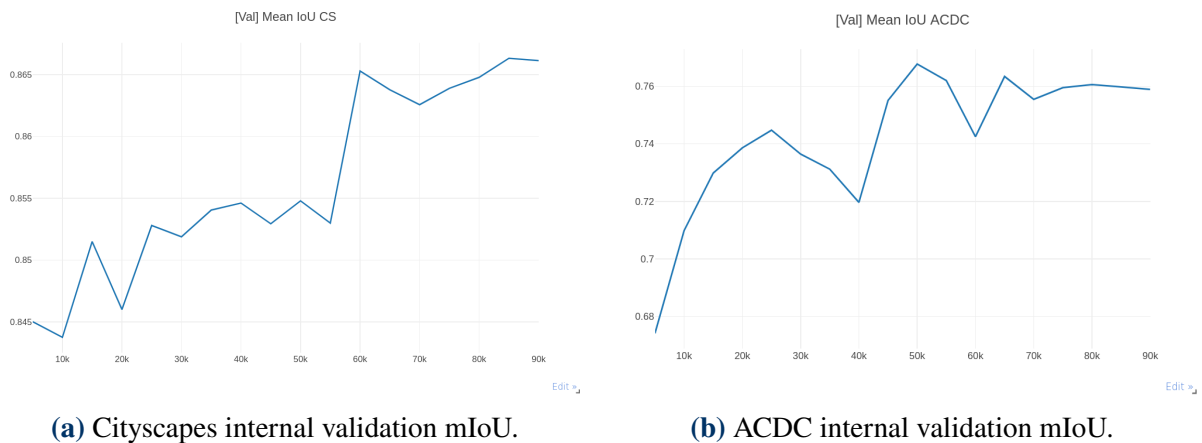


Figure 5.1: Internal validation mIoU curves monitored during training (Visdom).

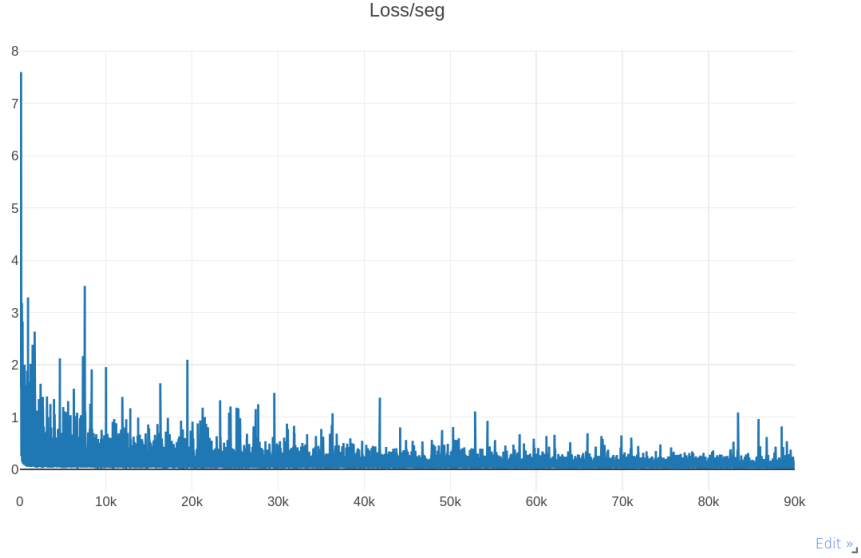


Figure 5.2: Segmentation loss curve monitored during training (Visdom).

5.8 Reproducibility and Logging

To support reproducibility, the 85/15 train/validation split for both datasets is generated deterministically using a fixed holdout seed (`holdout_seed = 1`). All configuration files (datasets, models, optimisers and schedules) are stored under version control, and training logs record loss curves, learning-rate values and periodic validation metrics. The conda environment (`mask2former-adverse`) is exported to `environment.yml` and captures the dependency versions used. Checkpoints are saved at regular intervals (every 5 000 iterations) with metadata indicating the run configuration.

5.9 Summary

This chapter detailed the hardware, software, preprocessing, training and evaluation setup used in this thesis. Key aspects are the high-performance workstation with an RTX 4090 GPU, real-data-only preprocessing for Cityscapes and ACDC, different crop and batch sizes for DeepLabV3+ and Mask2Former, and the alternate-batch training scheme with encoder freezing and WAS/TAS auxiliary supervision. Within each model family, all three training variants share the same core hyperparameters; only the training strategy changes. The next chapter presents the quantitative and qualitative results under this setup and analyses robustness to adverse weather and preservation of clear-weather performance.

6 Results and Discussion

6.1 Overview

This chapter evaluates the proposed semantic segmentation framework on both clear-weather (Cityscapes) and adverse-weather (ACDC) conditions using the **held-out evaluation sets (official validation splits) defined in Chapter 3**: the official Cityscapes validation split (500 images) and the official ACDC validation split (406 images), both treated as held-out evaluation sets in this thesis [2], [3]. The analysis focuses on two backbone families:

- **Baseline 1: DeepLabV3+** (CNN encoder–decoder with MobileNetV2 backbone) [1], [31],
- **Baseline 2: original Mask2Former** (transformer-based model with a Swin backbone) [4], [5].

For each backbone, three main training styles are compared:

1. **Trained on Cityscapes only:** the model is trained only on Cityscapes images and evaluated on both held-out evaluation sets (official validation sets).
2. **Alternate-batch training on Cityscapes and ACDC with selective encoder freezing:** training alternates between Cityscapes and ACDC mini-batches; the encoder is frozen on Cityscapes batches and trainable on ACDC batches.
3. **Alternate-batch training on Cityscapes and ACDC with selective encoder freezing and WAS/TAS:** same alternate-batch schedule and encoder freezing as in (2), with Weather-Aware Supervision (WAS) and Time-Aware Supervision (TAS) heads active during training.

Additional variants—including mixed-dataset training where CS and ACDC are combined as a single dataset—are introduced later in the ablation section (Section 6.7).

Results are first compared at an overall level (clear vs. adverse weather), then broken down per weather condition on ACDC (rain, fog, snow and night). For each case, the structure is:

- a summary table with mIoU on the held-out evaluation set,
- a bar chart comparing DeepLabV3+ and Mask2Former under the three training styles,
- a discussion of the trends visible in the plots.

Throughout this chapter, numerical tables report mIoU values on the standard $[0, 1]$ scale, whereas the bar charts display the same scores in percent (for example, 0.77 corresponds to 77 %). This choice improves readability while keeping the underlying numbers directly comparable.

This is followed by per-weather ACDC results, ablation studies across all stages, a runtime comparison, a dedicated ablation on encoder freezing and alternation schedule, and qualitative visualisations. Overall, the experiments show that the proposed training strategy substantially improves performance under adverse weather while maintaining strong performance on Cityscapes, which was the main goal of this thesis.

6.2 Baseline Models

Baseline 1 – DeepLabV3+

Baseline 1 is DeepLabV3+ with a MobileNetV2 backbone, following Kerim et al. [7]. This configuration is chosen because it forms the core model in the 2022 DeepLabV3+ adverse-weather framework and represents a strong, lightweight CNN baseline on Cityscapes [1]. Using it here allows a direct comparison between:

- the original CNN-based setup from Kerim et al., and
- the transformer-based Mask2Former framework proposed in this thesis,

under the same training styles and datasets.

Table 6.1 summarises the three DeepLabV3+ variants. The model architecture remains identical across all rows; only the training data and training strategy change.

Model	Training data	Training strategy
DeepLabV3+	CS	• trained on Cityscapes only; • standard DeepLabV3+ schedule from Kerim et al. [7]
DeepLabV3+	CS+ACDC	• alternate-batch training on CS and ACDC; • selective encoder freezing on CS batches; • encoder trainable on ACDC batches; • no WAS/TAS heads.
DeepLabV3+	CS+ACDC	• same alternate-batch training on CS and ACDC; • same selective encoder freezing as above; • WAS/TAS heads active during training.

Table 6.1: Baseline 1 (DeepLabV3+) training styles. “CS” denotes Cityscapes, “ACDC” denotes the adverse-weather dataset.

Baseline 2 – Original Mask2Former

Baseline 2 is the original Mask2Former model with a Swin backbone, implemented in the same codebase and framework described in Chapter 5 and following the original Mask2Former formulation [4], [5]. This architecture serves as the reference transformer backbone:

- it provides a strong clear-weather baseline on Cityscapes,
- it allows isolating the effect of the proposed training strategy (alternate-batch training, encoder freezing, WAS/TAS) by comparing the original Mask2Former configuration against the final model.

Table 6.2 lists the three Mask2Former variants. Again, the architecture remains fixed; only the training data and training strategy differ.

Model	Training data	Training strategy
Mask2Former	CS	• original Mask2Former training on Cityscapes only; • training configuration as described in Chapter 5 (without ACDC batches).
Mask2Former	CS+ACDC	• alternate-batch training on CS and ACDC; • Swin encoder frozen on CS batches; • encoder trainable on ACDC batches; • no WAS/TAS heads.
Mask2Former	CS+ACDC	• same alternate-batch schedule on CS and ACDC; • same selective encoder freezing as above; • WAS/TAS heads active during training.

Table 6.2: Baseline 2 (original Mask2Former) training styles. “CS” denotes Cityscapes, “ACDC” denotes the adverse-weather dataset.

These two baselines allow two types of comparison:

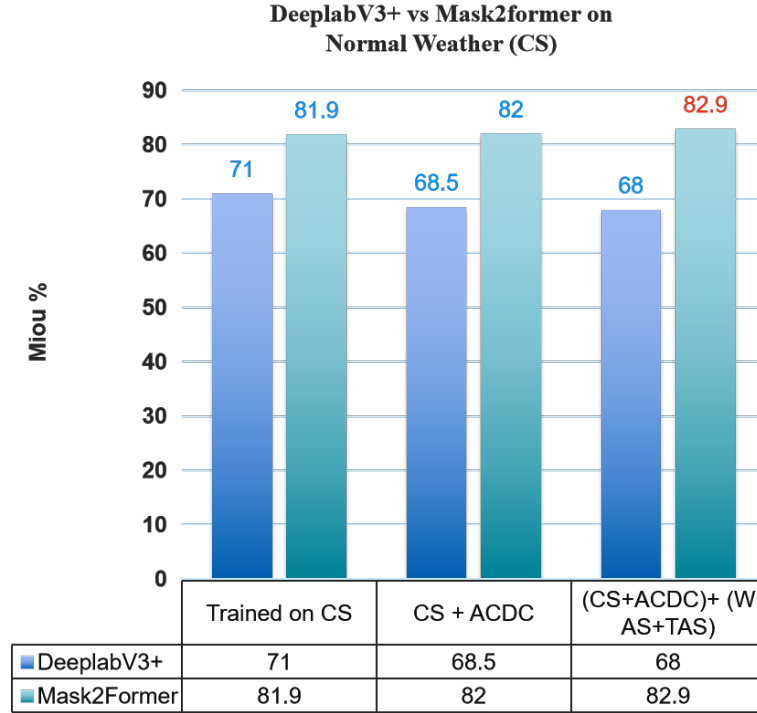
1. **Architecture comparison:** DeepLabV3+ vs. Mask2Former under the same training styles and datasets.
2. **Training-strategy comparison:** original Mask2Former vs. the proposed transformer framework with alternate-batch training, encoder freezing and WAS/TAS, plus additional variants without encoder freezing (Section 6.7).

6.3 Quantitative Results on the Cityscapes Official Validation Set

This section compares mean Intersection over Union (mIoU) on the **Cityscapes official validation set** (500 images, 19 classes) [2]. DeepLabV3+ and Mask2Former are evaluated under the three training styles introduced above. Table 6.3 reports the mIoU, and Figure 6.1 visualises the same numbers as a bar chart for easier comparison.

Table 6.3: Performance on the Cityscapes official validation set (mIoU, 19 classes).

Model	CS only	CS+ACDC (alt. + freeze)	CS+ACDC (alt. + freeze + WAS/TAS)
DeepLabV3+	0.71	0.685	0.68
Mask2Former	0.819	0.820	0.829

**Figure 6.1:** Overall mIoU on the Cityscapes official validation set for DeepLabV3+ and Mask2Former under the three training styles. Bars show mIoU in percent (e.g. 0.83 corresponds to 83 %).

Several trends are visible in Table 6.3 and Figure 6.1:

- **Transformer vs. CNN on clear weather.** With Cityscapes-only training, Mask2Former reaches about 81.9 % mIoU, while DeepLabV3+ is around 71 %. The ~10 percentage point gap confirms that the Swin-based backbone is a much stronger clear-weather baseline than MobileNetV2.
- **Effect of alternate CS+ACDC training.** Adding ACDC with alternate-batch training and selective encoder freezing slightly reduces DeepLabV3+ (71 % → 68.5 %), whereas Mask2Former essentially preserves its Cityscapes performance (81.9 % → 82.0 %), showing that the transformer backbone is more stable under multi-domain training.
- **Full proposed method.** With WAS/TAS enabled, DeepLabV3+ stays around 68 % on Cityscapes, while Mask2Former improves to 82.9 %. Thus, the proposed training strategy maintains—and even slightly improves—clear-weather performance for Mask2Former, whereas DeepLabV3+ trades a few points to gain robustness on ACDC.

To understand where these differences arise, it is useful to compare per-class performance. Table 6.4 therefore reports the per-class mIoU on the Cityscapes official validation set for the *final* training style (CS+ACDC with alternate-batch training, selective encoder freezing and WAS/TAS), comparing DeepLabV3+ and Mask2Former. For each semantic class, the better of the two models is shown in bold, and the last row reports the overall 19-class mIoU.

Table 6.4: Per-class mIoU on the Cityscapes official validation set for the final training style (CS+ACDC, alternate-batch + encoder freezing + WAS/TAS). Best results per class are shown in bold.

Class	DeepLabV3+	Mask2Former
Road	0.973	0.986
Sidewalk	0.800	0.878
Building	0.904	0.941
Wall	0.436	0.665
Fence	0.532	0.703
Pole	0.567	0.724
Traffic light	0.574	0.753
Traffic sign	0.707	0.841
Vegetation	0.906	0.932
Terrain	0.552	0.647
Sky	0.932	0.957
Person	0.740	0.861
Rider	0.452	0.693
Car	0.926	0.961
Truck	0.551	0.872
Bus	0.672	0.932
Train	0.617	0.864
Motorcycle	0.396	0.741
Bicycle	0.696	0.816
Overall (19-class)	0.681	0.829

On Cityscapes, Mask2Former outperforms DeepLabV3+ *for every single class*, not only in the overall mIoU. The gains are particularly large for thin and small structures such as wall, fence, pole, traffic light and traffic sign, as well as for dynamic objects such as rider, motorcycle and bicycle. For dominant classes such as road, sidewalk, building, vegetation, sky and car, the transformer backbone also yields consistently higher IoU than the CNN baseline. This confirms that the proposed training strategy combined with a strong transformer encoder not only preserves clear-weather performance, but improves both large-scale layout and fine-detail segmentation.

From a deployment perspective, this is critical: robustness in adverse weather must not come at the expense of performance in the clear-weather conditions that dominate real driving.

6.4 Quantitative Results on the ACDC Official Validation Set

We now evaluate on the full ACDC official validation set (406 images across rain, fog, snow and night) [3]. Table 6.5 reports overall mIoU, and Figure 6.2 (Figure 6.2) shows the same comparison as a bar chart.

Table 6.5: Performance on the ACDC official validation set (mIoU, 19 classes).

Model	CS only	CS+ACDC (alt. + freeze)	CS+ACDC (alt. + freeze + WAS/TAS)
DeepLabV3+	0.30	0.62	0.62
Mask2Former	0.63	0.762	0.772

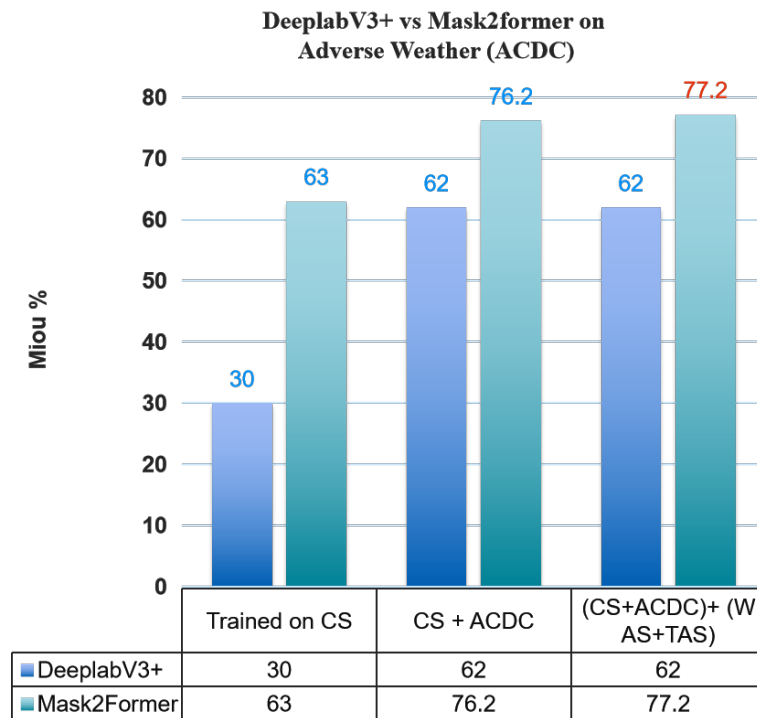


Figure 6.2: Overall mIoU on the ACDC official validation set for DeepLabV3+ and Mask2Former under the three training styles. Bars show mIoU in percent.

While Table 6.5 summarises the overall robustness on the ACDC official validation set, it does not reveal how performance is distributed across semantic classes. Table 6.6 therefore reports per-class mIoU on the ACDC official validation set for the final training style (CS+ACDC with alternate-batch training, encoder freezing and WAS/TAS), again comparing DeepLabV3+ and Mask2Former. As before, the better value for each class is shown in bold, and the last row reports the overall 19-class mIoU.

On ACDC, Mask2Former again dominates DeepLabV3+ across all 19 classes. The transformer backbone is particularly advantageous for dynamic and safety-critical objects (person, rider, car, bus, train and motorcycle), where the per-class IoU gap is large, but it also improves static structures such as road, sidewalk, wall, fence and vegetation. Even in challenging categories such

Table 6.6: Per-class mIoU on the ACDC official validation set for the final training style (CS+ACDC, alternate-batch + encoder freezing + WAS/TAS). Best results per class are shown in bold.

Class	DeepLabV3+	Mask2Former
Road	0.919	0.966
Sidewalk	0.694	0.839
Building	0.839	0.900
Wall	0.493	0.662
Fence	0.410	0.592
Pole	0.572	0.729
Traffic light	0.693	0.845
Traffic sign	0.576	0.724
Vegetation	0.857	0.890
Terrain	0.444	0.532
Sky	0.950	0.961
Person	0.570	0.745
Rider	0.195	0.557
Car	0.814	0.885
Truck	0.483	0.710
Bus	0.738	0.965
Train	0.866	0.930
Motorcycle	0.312	0.573
Bicycle	0.365	0.662
Overall (19-class)	0.621	0.772

as rider and motorcycle, where both models struggle, the proposed transformer-based framework achieves substantially higher IoU. The overall improvement from 0.621 to 0.772 mIoU shows that these gains are consistent across the entire class set, not driven by a single category.

Key observations:

- **CS-only training fails under domain shift.** When trained only on Cityscapes, DeepLabV3+ collapses to about 30 % mIoU on ACDC, while Mask2Former still manages around 63 %. This highlights both the severity of the domain gap and the advantage of the transformer backbone.
- **Alternate CS+ACDC training is essential.** Alternate-batch training with encoder freezing lifts DeepLabV3+ from 30 % to about 62 % and Mask2Former from 63 % to roughly 76.2 %. The bulk of robustness therefore comes simply from exposing the model to adverse-weather images in a controlled schedule. The additional ablation with mixed-dataset training (Stage 5 in Table 6.8) shows that Mask2Former reaches 0.767 ACDC overall and 0.827 on Cityscapes, i.e. comparable to alternate-batch training. This confirms that multi-domain CS+ACDC training itself is the dominant factor.
- **WAS/TAS provide an additional gain.** With WAS/TAS enabled, Mask2Former improves further to 77.2 % on ACDC, while DeepLabV3+ remains roughly unchanged around

62 %. The auxiliary tasks therefore act as a regulariser on top of the dominant effect of multi-domain training.

Taken together with the Cityscapes official validation set results, these numbers show that Mask2Former with the proposed training strategy attains strong robustness in adverse weather without sacrificing performance in clear weather. DeepLabV3+ cannot reach the same level even under the same training regime.

6.5 Per-Weather ACDC Evaluation

To understand the behaviour in different adverse conditions, the ACDC official validation set results are broken down into rain, fog, snow and night. Table 6.7 reports the per-weather mIoU for the main variants.

Table 6.7: Per-weather ACDC official validation set performance (mIoU, 19 classes).

Model / training style	Rain	Fog	Snow	Night
DeepLabV3+ (CS only)	0.35	0.47	0.30	0.11
DeepLabV3+ (CS+ACDC, alt. + freeze)	0.62	0.73	0.62	0.47
DeepLabV3+ (CS+ACDC, alt. + freeze + WAS/TAS)	0.62	0.73	0.625	0.48
Mask2Former (CS only)	0.69	0.80	0.63	0.37
Mask2Former (CS+ACDC, alt. + freeze)	0.803	0.816	0.798	0.554
Proposed Mask2Former (CS+ACDC, alt. + freeze + WAS/TAS)	0.812	0.825	0.804	0.578

Weather-wise observations:

- **Rain and snow.** DeepLabV3+ improves substantially once ACDC is added, but even then lags behind Mask2Former. The proposed method pushes Mask2Former above 0.80 mIoU in both rain and snow, showing that the model can handle reflections, water streaks and partially covered surfaces while preserving structure.
- **Fog.** Fog introduces strong haze and low contrast. The jump from 0.80 (Mask2Former CS-only) to 0.825 with the proposed method indicates that explicit alternate training and auxiliary supervision help recover features that would otherwise be washed out [9], [21].
- **Night.** Night scenes remain the most challenging condition: DeepLabV3+ reaches only 0.11 mIoU from CS-only training and 0.47 after adaptation, while Mask2Former with the full method reaches 0.578. The large gap, combined with the TAS head supervising day vs. night, suggests that the transformer backbone plus time-aware supervision is particularly effective for low-light scenes.

These per-condition results confirm that the proposed design brings consistent gains across all adverse scenarios, not only in the aggregate ACDC score.

6.6 Effect of Training Style per Weather Condition

The overall and per-weather tables provide a numerical summary. To better understand how each backbone behaves across the different weather conditions, it is useful to inspect the per-weather mIoU in a single plot for each model family. This section therefore discusses the training-style bar charts for DeepLabV3+ and Mask2Former.

6.6.1 Baseline 1 – DeepLabV3+

For DeepLabV3+, the impact of the three training styles is illustrated in Figure 6.3, which reports mIoU for Cityscapes and for each ACDC subset (rain, fog, snow, night). Each group of bars corresponds to one training style from Table 6.1.

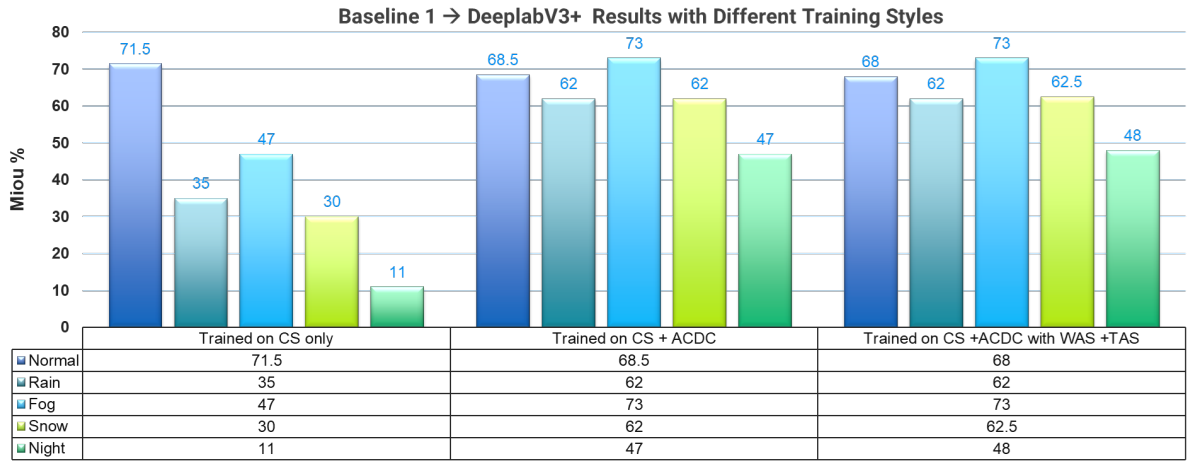


Figure 6.3: DeepLabV3+ results with different training styles. Each group of bars corresponds to one training style and shows mIoU (in percent) for Cityscapes (clear weather) and the four ACDC subsets.

DeepLabV3+ trained on Cityscapes only. With CS-only training, DeepLabV3+ performs reasonably well on Cityscapes (≈ 71 % mIoU) but generalises very poorly to ACDC: rain and fog are already weak, snow is worse and night scenes are close to unusable (≈ 11 % mIoU). The CNN baseline therefore has almost no intrinsic robustness to adverse weather.

DeepLabV3+ with alternate CS+ACDC training and encoder freezing. Alternate-batch training with encoder freezing changes the picture. Cityscapes performance drops slightly (71 % $\rightarrow 68.5$ %), but all ACDC subsets jump into the 60 – 70 % range and night scenes improve from about 11 % to roughly 47 %. A modest sacrifice in clear-weather performance thus buys a massive gain under adverse conditions.

DeepLabV3+ with full proposed training (WAS/TAS). Adding WAS/TAS on top of the alternate schedule does not radically change DeepLabV3+. Clear-weather performance remains

around 68 %, and the adverse metrics stay close to those of the previous variant, with a small but consistent improvement at night (47 % $\rightarrow$ 48 %). For the CNN baseline, the dominant effect clearly comes from adding ACDC and using the alternate schedule; WAS/TAS act mainly as a mild refinement.

6.6.2 Baseline 2 – Original Mask2Former

For Mask2Former, the corresponding per-weather behaviour is shown in Figure 6.4, again reporting mIoU on the Cityscapes and ACDC held-out evaluation sets (official validation splits) for the three training styles in Table 6.2.

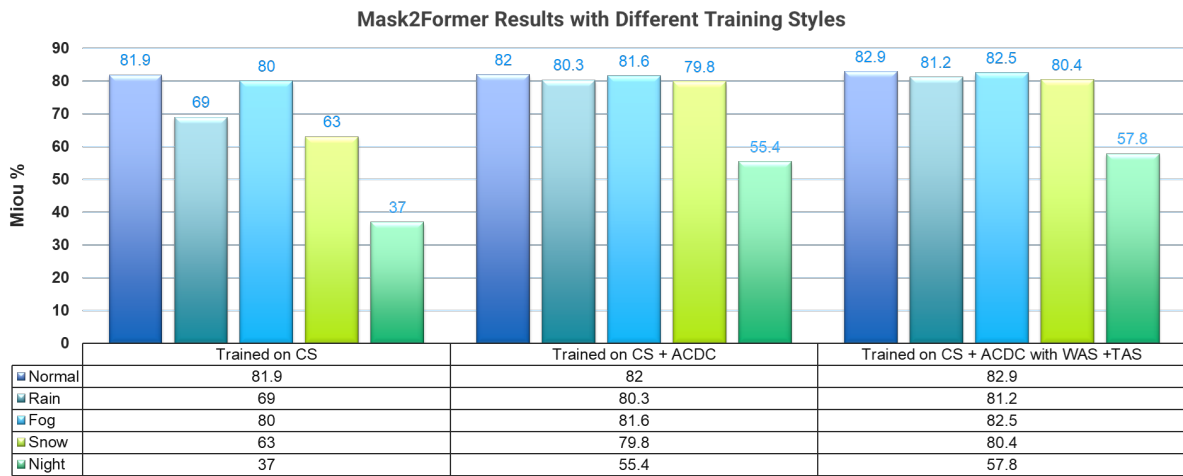


Figure 6.4: Mask2Former results with different training styles. Each group of bars shows mIoU (in percent) on Cityscapes (“Normal”) and the four ACDC weather subsets.

Mask2Former trained on Cityscapes only. With CS-only training, Mask2Former already achieves 81.9 % mIoU on Cityscapes and surprisingly strong performance in rain, fog and snow, but night scenes remain clearly weaker (about 37 % mIoU). The transformer backbone is therefore more robust than DeepLabV3+ even without seeing adverse-weather images, but there is still significant room for improvement.

Mask2Former with alternate CS+ACDC training and encoder freezing. When alternate-batch training and selective encoder freezing are enabled, Cityscapes performance does *not* degrade; it slightly increases to around 82 %. At the same time, all adverse-weather conditions improve strongly: rain moves into the low 80 % range, snow jumps from the low 60s to almost 80 %, and night scenes increase from roughly 37 % to more than 55 %. From this comparison alone, it appears that most of the robustness gain is tied to the multi-domain CS+ACDC training. The additional ablation in Section 6.7 (Stage 5) shows that mixed-dataset training (treating CS and ACDC as one combined dataset) achieves 0.767 ACDC overall and 0.827 on Cityscapes, confirming that joint training is the dominant factor, with differences of at most about two percentage points per weather subset compared to alternate-batch training.

Mask2Former with full proposed training (WAS/TAS). With WAS/TAS branches active, Mask2Former improves again across all conditions, although the gains are now smaller because the model is already strong. Cityscapes mIoU rises to about 82.9 %, and all adverse-weather subsets gain another one to two percentage points. The effect is most visible at night (55.4 % → 57.8 %), matching the intuition that TAS is particularly relevant when illumination varies strongly.

Overall, Baseline 2 demonstrates that:

- the original Mask2Former already has decent robustness to adverse conditions,
- multi-domain CS+ACDC training provides the largest robustness boost, while encoder freezing behaves mainly as a mild regulariser rather than an essential component (Section 6.7),
- WAS/TAS offer a consistent additional gain, especially for night scenes.

6.7 Ablation Studies

The previous sections examined results along two axes: backbone choice (DeepLabV3+ vs. Mask2Former) and training style. This section consolidates these experiments into a single ablation table, introduces an additional ablation on encoder freezing and alternation schedule, and discusses the incremental effect of each stage.

6.7.1 Consolidated Ablation Table

For compactness, Table 6.8 uses numerical stage identifiers 1–7:

- **Stage 1:** DeepLabV3+ trained on CS only.
- **Stage 2:** DeepLabV3+ trained on CS+ACDC with alternate batches and selective encoder freezing.
- **Stage 3:** DeepLabV3+ trained on CS+ACDC with alternate batches, selective encoder freezing and WAS/TAS.
- **Stage 4:** Mask2Former trained on CS only.
- **Stage 5:** Mask2Former trained on a mixed dataset (CS+ACDC combined as one dataset) with a fully trainable encoder (no WAS/TAS).
- **Stage 6:** Mask2Former trained on CS+ACDC with alternate batches and selective encoder freezing (no WAS/TAS).
- **Stage 7:** Mask2Former trained on CS+ACDC with alternate batches, selective encoder freezing and WAS/TAS (final proposed model).

The table reports per-weather mIoU on the ACDC official validation set as well as overall ACDC and Cityscapes mIoU.

Stage	Rain	Fog	Snow	Night	ACDC overall	Cityscapes overall
1	0.35	0.47	0.30	0.11	0.30	0.71
2	0.62	0.73	0.62	0.47	0.62	0.68
3	0.62	0.73	0.625	0.48	0.62	0.68
4	0.69	0.80	0.63	0.37	0.63	0.819
5	0.807	0.829	0.809	0.56	0.767	0.827
6	0.803	0.816	0.798	0.554	0.762	0.820
7	0.812	0.825	0.804	0.578	0.772	0.829

Table 6.8: Consolidated ablation results across stages 1–7 (mIoU) on the Cityscapes and ACDC official validation sets.

6.7.2 Additional Ablation: Mixed-Dataset Training vs. Alternate-Batch Training

Stage 5 implements mixed-dataset training where Cityscapes and ACDC are combined as a single dataset, while Stage 6 uses alternate-batch training with encoder freezing. Both are trained without WAS/TAS.

The two stages yield comparable performance: ACDC overall is 0.767 (Stage 5) vs. 0.762 (Stage 6), and Cityscapes overall is slightly higher for Stage 5 (0.827 vs. 0.820). Per-weather differences remain within approximately two percentage points. This confirms that for Mask2Former, the core factor is joint CS+ACDC training itself; the specific implementation—whether mixed-dataset or alternate-batch with freezing—has only a secondary, regularising effect.

6.7.3 Ablation Discussion

Reading Table 6.8 stage by stage:

- **Stage 1 → Stage 2.** Moving from DeepLabV3+ on CS only to alternate CS+ACDC training with encoder freezing more than doubles ACDC overall mIoU ($0.30 \rightarrow 0.62$), at the cost of a modest drop on Cityscapes ($0.71 \rightarrow 0.68$). This confirms that joint CS+ACDC training under the chosen alternation schedule is crucial even for the CNN baseline.
- **Stage 2 → Stage 3.** Adding WAS/TAS on DeepLabV3+ does not change the numbers dramatically; the model appears capacity-limited, so most of the gain comes from simply seeing ACDC at all.

- **Stage 1 → Stage 4.** Replacing DeepLabV3+ by Mask2Former while still training on CS only already yields a huge improvement on ACDC overall (0.30 → 0.63) and a higher Cityscapes score (0.819 vs. 0.71). This directly validates the first design objective: using a stronger transformer backbone.
- **Stage 4 → Stage 5.** Introducing joint CS+ACDC training via mixed-dataset training (Stage 5) increases ACDC overall from 0.63 to 0.767, while also improving Cityscapes from 0.819 to 0.827. This demonstrates that multi-domain training provides a substantial robustness gain without any encoder freezing or auxiliary tasks.
- **Stage 5 → Stage 6.** Switching from mixed-dataset training (Stage 5) to alternate-batch training with encoder freezing (Stage 6) results in slightly lower ACDC overall (0.767 vs. 0.762) and Cityscapes performance (0.827 vs. 0.820). This indicates that mixed-dataset training performs comparably to, or even slightly better than, the more structured alternate-batch approach with encoder freezing.
- **Stage 6 → Stage 7.** Enabling WAS/TAS on top of the alternate schedule and encoder freezing leads to the final model with 0.772 ACDC overall and 0.829 on Cityscapes. Relative to Stage 4 (CS-only Mask2Former), this is an absolute gain of $0.772 - 0.63 = 0.142$ on ACDC, while also improving clear-weather performance (0.819 → 0.829). The biggest relative improvement is observed at night (0.37 → 0.578).

Overall, the ablation shows that three elements clearly dominate the performance:

1. adopting a stronger transformer backbone (Mask2Former),
2. using multi-domain CS+ACDC training, and
3. adding WAS/TAS auxiliary tasks.

Encoder freezing and the precise alternation schedule have only a secondary impact: Stage 5 and Stage 6 demonstrate that simpler or more structured schedules both work, as long as CS and ACDC are trained jointly. Stage 7 then builds on this with WAS/TAS to yield the best overall balance between clear-weather performance and robustness in adverse conditions.

6.8 Runtime and Inference Speed

The previous sections focused on segmentation quality in terms of mIoU on Cityscapes and ACDC (Sections 6.3– 6.5). For practical deployment, however, inference speed is equally important. This section therefore reports a concise runtime comparison between the DeepLabV3+ baseline and the proposed Mask2Former-based model.

All timings were measured on the same workstation (Ubuntu 22.04) with a single NVIDIA GeForce RTX 4090 GPU (24 GB VRAM), using the same software stack as in Chapter 5 (PyTorch build and CUDA runtime). Unless stated otherwise, the evaluation batch size is 3.

Note that the two models are benchmarked under the crop sizes used in this thesis (DeepLabV3+: 768×768 , Mask2Former: 640×640), so the runtime comparison should be interpreted as indicative rather than resolution-matched.

Pure inference time per image

To measure the actual network inference cost, both models were evaluated with an internal benchmark mode that:

- runs a short warm-up phase,
- then measures only the forward pass (no metrics, no logging, no saving),
- and reports the average wall-clock time per image.

For the final checkpoints used in this thesis (DeepLabV3+ and Mask2Former both trained on CS+ACDC with alternate batches, encoder freezing and WAS/TAS), the benchmark results are:

- **DeepLabV3+ (MobileNetV2, crop 768×768):**
 - Cityscapes official validation set: **14.374 ms** per image (≈ 69.6 FPS),
 - ACDC official validation set: **14.322 ms** per image (≈ 69.8 FPS).

In practice, DeepLabV3+ runs at about **14.3 ms per image** on the RTX 4090, i.e. roughly **70 FPS**.

- **Mask2Former (Swin-based, crop 640×640):**
 - Cityscapes official validation set: **605.503 ms** per image (≈ 1.65 FPS),
 - ACDC official validation set: **594.125 ms** per image (≈ 1.68 FPS).

Thus, Mask2Former requires about **0.60 s per image**, corresponding to only **1.7 FPS**.

Overall, the transformer-based model is roughly 40–42× **slower in pure forward-pass time** than the lightweight DeepLabV3+ baseline on the same hardware.

End-to-end evaluation runtime: DeepLabV3+

In addition to the pure inference benchmark, it is useful to consider the total time for a full evaluation pass (prediction *plus* metric computation). Table 6.9 summarises the end-to-end evaluation time for DeepLabV3+ with a MobileNetV2 backbone (output stride 16, crop size 768×768 , batch size 3).

Even when including metric computation, DeepLabV3+ evaluates the full Cityscapes official validation set in well under a minute and the ACDC official validation set in roughly 23 s. On the evaluation workstation, this is near real-time for many perception pipelines.

Table 6.9: End-to-end evaluation runtime for DeepLabV3+ (MobileNetV2, batch size 3), including metric computation, on the official validation sets (held-out evaluation sets).

Dataset	#Images	Total time	Approx. time / image
Cityscapes official validation set	500	≈ 43 s	≈ 86 ms
ACDC official validation set	406	≈ 23 s	≈ 57 ms

End-to-end evaluation runtime: Mask2Former

Table 6.10 reports the corresponding end-to-end evaluation runtimes for the proposed Mask2Former-based model with a Swin backbone (crop size 640×640 , batch size 3).

Table 6.10: End-to-end evaluation runtime for Mask2Former (Swin-based, batch size 3), including metric computation, on the official validation sets (held-out evaluation sets).

Dataset	#Images	Total time	Approx. time / image
Cityscapes official validation set	500	≈ 16.5 min	≈ 2.0 s
ACDC official validation set	406	≈ 10.2 min	≈ 1.5 s

Compared to DeepLabV3+, Mask2Former requires approximately:

- $\sim 23\times$ more time per image on Cityscapes (from ≈ 86 ms to ≈ 2.0 s),
- $\sim 26\times$ more time per image on ACDC (from ≈ 57 ms to ≈ 1.5 s).

The end-to-end slowdown is smaller than the pure forward-pass slowdown ($\sim 40\times$), because metric computation and data-loading overheads constitute a fixed cost that affects the faster model more strongly in relative terms.

Interpretation and Architectural Trade-offs

Combining these runtime measurements with the mIoU results in Sections 6.3 and 6.4 leads to a clear trade-off:

- **DeepLabV3+** offers competitive segmentation quality at a very low inference cost: around **14.3 ms per image** in pure forward pass. It can therefore operate close to real-time on a modern GPU.
- **Mask2Former** delivers substantially higher mIoU, especially on ACDC and at night, but at the cost of a significant increase in per-image processing time: roughly **0.6 s per image** in pure inference. In its current configuration, it is more suitable for offline processing or high-end hardware.

Architectural Reasons for the Performance Trade-off. The substantial performance gap between Mask2Former and DeepLabV3+ stems from fundamental architectural differences.

Mask2Former leverages transformer-based self-attention mechanisms that model *long-range dependencies* across the entire image, allowing it to better understand global context—especially valuable in adverse conditions. This comes at the cost of higher computational demand. In contrast, DeepLabV3+ uses convolutional operations that are *locally restricted* and highly optimized for GPU execution, making them faster but less capable of capturing global scene structure.

Potential Optimization Strategies for Real-Time Deployment. Several techniques could bridge the accuracy-speed gap for practical applications:

- **Model Compression:** Pruning or knowledge distillation to smaller backbones.
- **Quantization:** Mixed-precision (FP16/INT8) inference.
- **Architectural Modifications:** Efficient attention variants or hybrid designs.
- **Hardware-Specific Optimization:** TensorRT for automotive GPUs.
- **Multi-Model Cascades:** Switching to Mask2Former only in adverse conditions.

These strategies could reduce Mask2Former’s inference time by 3–10× while preserving most of its accuracy advantages.

Which model is preferable depends on the target use case: for research and high-accuracy offline evaluation, the transformer-based Mask2Former is preferable; for tight runtime budgets and embedded deployment, a lighter backbone such as DeepLabV3+ remains more practical. A reasonable engineering compromise is to treat transformer segmentation as a “robust mode” for adverse conditions.

6.8.1 Inference Output Visualization

Complementing the quantitative runtime analysis in the previous section, Figure 6.5 provides a comprehensive visual summary of the artifacts generated during a single inference pass of the proposed model. The figure systematically demonstrates the complete output pipeline for representative examples across all weather conditions studied in this thesis. For each of the four adverse weather conditions in the ACDC dataset (rain, fog, snow, and night) and one clear-weather (‘normal’) condition from the Cityscapes dataset, the visualization shows the corresponding outputs produced by the final proposed Mask2Former model (Stage 7).

This visualization serves two primary purposes: (1) to illustrate the types of visual artifacts that are automatically generated during inference for any input image, and (2) to provide qualitative insight into how the model’s predictions manifest across different challenging conditions. The grid format allows for direct comparison between input images and their corresponding outputs, highlighting both the consistency and variations in prediction quality across weather domains.

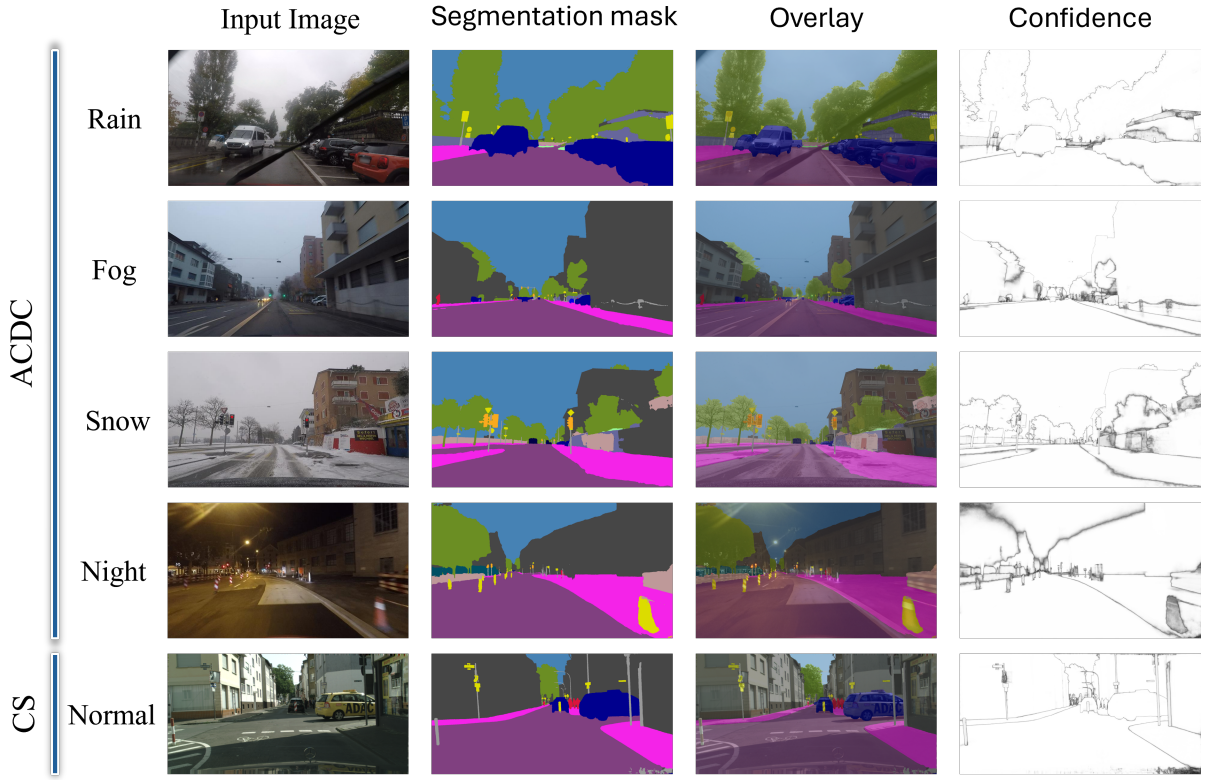


Figure 6.5: Visualization of standard inference outputs across weather conditions. For each sample input (first column), the model generates: the semantic segmentation mask (second column), the mask overlaid on the input image (third column), and a pixel-wise prediction confidence map (fourth column). The confidence map visualizes model certainty (brighter indicates higher confidence). A prediction ID map (assigning a unique instance identifier) is also produced but omitted here as it appears primarily black and is less informative for qualitative assessment. This figure illustrates the core visual results available after running inference on any input image.

For every input image, a standard inference run yields several outputs:

- **Segmentation Mask:** The primary semantic prediction using the standard Cityscapes 19-class colour palette.
- **Overlay:** The segmentation mask blended with the original input image, providing contextual interpretation.
- **Confidence Map:** A grayscale image where pixel intensity indicates the model’s confidence in its prediction for that location (brighter = higher confidence).
- **Prediction ID Map:** An output that assigns a unique identifier to each object instance. This map is typically used for instance-level analysis but is excluded from this visualization as it appears largely black and does not convey intuitive qualitative information.

This visualization confirms that the proposed framework not only meets quantitative performance goals but also produces a complete, interpretable set of outputs suitable for analysis and debugging in practical deployment scenarios.

6.9 Qualitative Results

Quantitative metrics summarise performance but do not show how predictions change visually. This section presents a visual comparison between both baselines and the final proposed implementation.

6.9.1 Visual Comparison Between Baselines and Final Approach

Figure 6.6 shows qualitative examples for four ACDC conditions (rain, snow, fog, night) and one Cityscapes scene (normal weather). For each input image, predictions from four models are compared:

1. DeepLabV3+ trained on Cityscapes only,
2. DeepLabV3+ with the proposed training style (CS+ACDC, alt. + freeze + WAS/TAS),
3. original Mask2Former trained on Cityscapes only,
4. proposed Mask2Former (CS+ACDC, alt. + freeze + WAS/TAS).

The ground-truth label map is shown in the right-most column. White dashed boxes highlight regions where the differences are most informative (e.g. footpaths, fences or side roads). All prediction maps use the same 19-class CS colour palette as introduced in the datasets chapter.

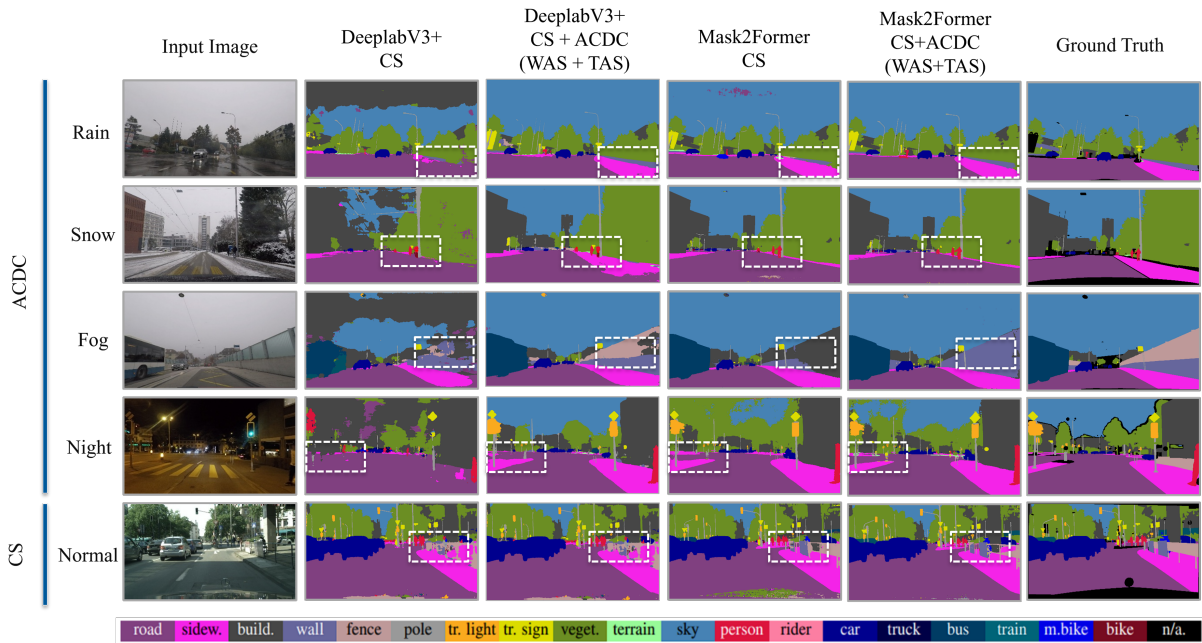


Figure 6.6: Visual comparison between baselines and the final proposed implementation across rain, snow, fog, night (ACDC) and normal (Cityscapes) conditions. Columns: input image, DeepLabV3+ (CS only), DeepLabV3+ (proposed training), original Mask2Former (CS only), proposed Mask2Former, ground truth. All predictions use the same Cityscapes colour palette.

Representative observations:

- **Rain.** In the rainy scene, CS-only DeepLabV3+ misses the footpath almost completely inside the highlighted box and also shows irregularities in the sky. With the proposed training, DeepLabV3+ recovers most of the footpath but boundaries remain fuzzy. Original Mask2Former improves sharpness, and the proposed Mask2Former both detects and delineates the footpath cleanly, very close to the ground truth.
- **Snow.** The snow example focuses on the footpath and three pedestrians. All models detect the pedestrians, but CS-only DeepLabV3+ fails to segment the footpath. DeepLabV3+ with the proposed training detects parts of it but with gaps and misalignments. Both Mask2Former variants are much sharper; the proposed model yields the most coherent separation between road, footpath and terrain despite snow coverage.
- **Fog.** In the foggy scene, the key structure is a wall with fencing on top. CS-only DeepLabV3+ largely misses this structure, while the proposed DeepLabV3+ detects it but with inconsistent labels. Original Mask2Former tends to merge the wall and fencing into the building class. The proposed Mask2Former captures the wall and fencing as a coherent structure, with boundaries that closely follow the ground truth.
- **Night.** Night-time images are the most challenging. CS-only DeepLabV3+ almost completely fails to separate the intersection: large areas of sidewalk are mislabelled as road. DeepLabV3+ with the proposed training recovers a rough road layout but still confuses road and sidewalk near the crossings. The original Mask2Former is already sharper, and the proposed Mask2Former further refines lane and sidewalk boundaries, aligning well with ground truth. This visually supports the strong numerical gains for night scenes in Tables 6.7 and 6.8.
- **Normal weather (Cityscapes).** In the Cityscapes example, all models perform reasonably well, but there are still differences. CS-only DeepLabV3+ and its extended variant struggle with fine structures such as the side fence. Mask2Former yields much cleaner fence geometry, and the proposed model preserves this quality despite being trained jointly on Cityscapes and ACDC.

These examples illustrate the main conclusions from the quantitative analysis: the proposed Mask2Former model preserves clear-weather quality while significantly improving robustness in adverse conditions.

6.10 Relation to Research Questions

The results can now be mapped back to the research questions from Chapter 1:

- **RQ1 – How well does a Cityscapes-pretrained model generalise to adverse weather?** Stage 4 (original Mask2Former, CS only) reaches 0.819 mIoU on the Cityscapes official validation set but only 0.63 mIoU on the ACDC official validation set overall, with night scenes dropping to 0.37 (Tables 6.5 and 6.7). This confirms a substantial domain gap between clear and adverse-weather conditions.
- **RQ2 – What is the effect of CS+ACDC joint training and encoder freezing?** Moving from Stage 4 to Stage 6 (Mask2Former, CS-only → CS+ACDC with alternate-batch training and encoder freezing) improves ACDC overall from 0.63 to 0.762, while Cityscapes performance remains essentially unchanged (0.819 → 0.820) (Table 6.8). The per-weather analysis in Table 6.7 and Figure 6.4 shows that this gain is consistent across rain, fog, snow and night. Stage 5 demonstrates that mixed-dataset training (treating CS and ACDC as one combined dataset) achieves 0.767 ACDC overall and 0.827 on Cityscapes, showing that joint multi-domain training is the dominant factor, while specific scheduling methods (alternate-batch vs. mixed-dataset) and encoder freezing are secondary considerations.
- **RQ3 – Does WAS/TAS auxiliary supervision help?** Adding WAS/TAS (Stage 6 → Stage 7) pushes Mask2Former from 0.762 to 0.772 ACDC overall and from 0.554 to 0.578 mIoU at night (Table 6.8). The qualitative grid (Figure 6.6) confirms that these improvements translate into sharper segmentation in low-light scenes. For DeepLabV3+, the gains are smaller but still positive, indicating that auxiliary tasks are most effective when the backbone has sufficient capacity.
- **RQ4 – How does the proposed transformer-based framework compare to a DeepLabV3+ baseline?** Across all training styles and weather conditions, Mask2Former outperforms DeepLabV3+ by a clear margin (Tables 6.3– 6.8 and Figures 6.1– 6.4). With the full proposed training (Stage 7), the gap is especially large on ACDC and at night, confirming that a transformer-based backbone is better suited for the multi-domain adverse-weather setting studied in this thesis.

6.11 Summary

This chapter presented a detailed quantitative and qualitative evaluation of the proposed training strategy on the Cityscapes and ACDC held-out evaluation sets (official validation sets). The main findings are:

- Cityscapes-pretrained models suffer from a strong performance drop in adverse weather, particularly at night, even when using a powerful backbone such as Mask2Former.
- Adding ACDC with a joint CS+ACDC training schedule is crucial to recover performance under adverse conditions. Both mixed-dataset training (Stage 5) and alternate-batch training with encoder freezing (Stage 6) yield comparable results, with mixed-dataset training showing slightly better ACDC overall performance (0.767 vs. 0.762). This confirms that the core requirement is joint training across domains, while specific implementation details have marginal impact.
- WAS/TAS auxiliary supervision provides an additional, consistent improvement, most notably for night scenes, and helps disentangle illumination and semantic content.
- The transformer-based Mask2Former backbone consistently outperforms the CNN-based DeepLabV3+ across all training styles and weather conditions.
- The final proposed model (Stage 7) reaches approximately 0.83 mIoU on Cityscapes and 0.77 mIoU on ACDC overall, corresponding to an absolute gain of about 14 percentage points on ACDC compared to the original Mask2Former trained on Cityscapes only (Stage 4).
- This accuracy gain comes at a significant runtime cost: on a single RTX 4090 GPU, Mask2Former requires on the order of 1.5–2.0 s per image for full evaluation, compared to tens of milliseconds for DeepLabV3+, which remains close to real-time. Architectural differences explain this trade-off: transformers capture global context through self-attention at higher computational cost, while CNNs offer efficient local processing.

The thesis achieves its core objective: designing a single semantic segmentation model that maintains high performance in clear-weather conditions while being significantly more robust in adverse conditions. The ablation studies confirm that the core ingredients are the transformer backbone, multi-domain CS+ACDC training and WAS/TAS supervision, while encoder freezing and the exact alternation schedule mainly act as optional regularisation choices. Chapter 7 discusses the implications of these results and outlines directions for future work, including extending the training strategy to synthetic adverse-weather datasets and to 3D object detection under adverse conditions.

7 Conclusion and Future Work

7.1 Conclusion

This thesis presents a comprehensive investigation of transformer-based semantic segmentation for autonomous driving applications in adverse weather conditions, utilizing exclusively real-world datasets—Cityscapes and ACDC [2], [3]. The primary research objective was to develop a *single* semantic segmentation model capable of reliable performance across both clear-weather (Cityscapes) and adverse-weather (ACDC) conditions, eliminating the necessity for synthetic data augmentation or specialized domain-specific models [7], [8], [9].

Experimental Protocol: During training, the official training splits of Cityscapes and ACDC were partitioned into internal 85% training and 15% validation subsets using a deterministic seed. Checkpoint selection was based on internal validation performance, with final results reported on the official validation splits (500 Cityscapes images, 406 ACDC images), which serve as held-out evaluation sets for this thesis; the Cityscapes official test set was not utilized.

Methodological Framework: The proposed approach builds upon a Mask2Former architecture with a Swin-L backbone pretrained on Cityscapes [2], [4], [5]. The methodology integrates four principal components:

- (i) Joint Cityscapes+ACDC optimisation
- (ii) Structured domain alternation
- (iii) Selective encoder freezing as an optional regulariser
- (iv) Lightweight WAS/TAS auxiliary heads with conservative loss weights ($\alpha = \beta = 10^{-5}$)

Key Experimental Findings:

- **Joint training emerges as the dominant factor** for achieving robustness across weather conditions while preserving clear-weather performance.
- The **final model (Stage 7)** achieves **0.83 mIoU** on Cityscapes and **0.77 mIoU** on ACDC, representing an absolute improvement of **14 percentage points** on ACDC compared to the Cityscapes-only Mask2Former baseline (Stage 4) (Table 6.8; Section 6.7).
- **WAS/TAS supervision** provides consistent incremental gains, particularly in night-time scenarios (Table 6.8; Section 6.7).
- Both **mixed-dataset training** and **alternate-batch training with encoder freezing** yield comparable robustness, indicating that the fundamental requirement is joint multi-domain training rather than specific scheduling methodology.

Research Questions Addressed:

- **RQ1:** A Cityscapes-pretrained model exhibits significant performance degradation on ACDC, confirming a substantial domain gap between clear and adverse conditions (Tables 6.3–6.5; Sections 6.3–6.5).
- **RQ2:** Joint Cityscapes+ACDC training is essential for cross-domain robustness; encoder freezing serves primarily as an optional regulariser rather than a critical component (Table 6.8; Section 6.7.2).
- **RQ3:** WAS/TAS auxiliary supervision yields measurable improvements, most pronounced in challenging low-light conditions (Table 6.8; Section 6.7.3).
- **RQ4:** The transformer-based Mask2Former framework consistently outperforms the CNN-based DeepLabV3+ (MobileNetV2) baseline across all training regimes, with substantial advantages under adverse weather evaluation (Tables 6.3, 6.5, and 6.8).

In summary, this work demonstrates that a single Mask2Former model can substantially bridge the performance gap between clear and adverse weather conditions using only real-world data, provided training incorporates both domains and auxiliary supervision is applied judiciously. This research contributes to the broader goal of developing robust perception systems for autonomous vehicles, showing that carefully designed training strategies can significantly mitigate performance degradation in adverse conditions without compromising clear-weather accuracy.

Limitations

Several limitations of this work warrant acknowledgment:

- **Dataset scope:** Evaluation is confined to Cityscapes and ACDC; generalization to other geographic regions, camera configurations, and weather distributions remains unverified.
- **Computational efficiency:** The Swin-L backbone imposes substantial computational demands, limiting real-time deployment on automotive hardware without optimization or adoption of more compact architectures [4], [5].
- **Sensor modality:** The approach operates exclusively on RGB imagery; multi-modal fusion (e.g., RGB-LiDAR or RGB-radar) lies beyond the scope of this investigation.
- **Hyperparameter exploration:** The explored hyperparameter and schedule space is necessarily limited; extended exploration may yield further improvements in the robustness-efficiency trade-off.

7.2 Future Work

Several directions are promising extensions of this work:

- **Synthetic augmentation:** Integrating synthetic adverse-weather data (e.g., AWSS) could enhance robustness while reducing reliance on manually annotated real adverse-weather data [7], [24], [26]. Systematic investigation of mixed real-synthetic training regimes would clarify the substitutive potential of synthetic data.
- **Domain adaptation beyond supervised joint training:** Feature-alignment methods such as adversarial adaptation or self-ensembling could reduce the domain gap further, especially for rare conditions where direct supervision is limited [21], [23], [25].
- **Anomaly and out-of-distribution awareness:** Integrating uncertainty estimation or anomaly segmentation could improve safety in rare or extreme conditions not represented in ACDC (unusual lighting, sensor faults, atypical infrastructure).
- **Real-time optimization:** Practical deployment necessitates efficiency gains via model compression, quantization, knowledge distillation, or adoption of efficient transformer variants.
- **Broader cross-dataset validation:** Testing on additional datasets (e.g., nuScenes) would better characterize generalization limits and dataset-specific failure modes.

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